

DOT OR SCALAR PRODUCT OF VECTORS



SYNOPSIS

➤ Let $\vec{a}$ and $\vec{b}$ be two nonzero vectors and θ be the angle between them, then the scalar product or dot product of $\vec{a}$ and $\vec{b}$ denoted by $\vec{a} \cdot \vec{b}$ is defined as $\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$

➤ If $(\vec{a}, \vec{b}) = \theta$, then $\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}$

➤ Let $\vec{i}, \vec{j}, \vec{k}$ be unit orthogonal vectors i.e. $|\vec{i}| = |\vec{j}| = |\vec{k}| = 1$ and $(\vec{i}, \vec{j}) = (\vec{j}, \vec{k}) = (\vec{k}, \vec{i}) = \frac{\pi}{2}$,

$$\text{then } \vec{i} \cdot \vec{i} = \vec{j} \cdot \vec{j} = \vec{k} \cdot \vec{k} = 1$$

$$\vec{i} \cdot \vec{j} = \vec{j} \cdot \vec{k} = \vec{k} \cdot \vec{i} = 0$$

$$\vec{j} \cdot \vec{i} = \vec{k} \cdot \vec{j} = \vec{i} \cdot \vec{k} = 0$$

➤ If $\vec{a} = a_1 \vec{i} + a_2 \vec{j} + a_3 \vec{k}$, $\vec{b} = b_1 \vec{i} + b_2 \vec{j} + b_3 \vec{k}$ Then

$$(i) \vec{a} \cdot \vec{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

$$(ii) \vec{a} \cdot \vec{a} = a_1^2 + a_2^2 + a_3^2 \quad (iii) \vec{a} \cdot \vec{a} = |\vec{a}|^2$$

➤ Let $\vec{a} = a_1 \vec{i} + a_2 \vec{j} + a_3 \vec{k}$, $\vec{b} = b_1 \vec{i} + b_2 \vec{j} + b_3 \vec{k}$ and let $(\vec{a}, \vec{b}) = \theta$ then

$$\cos \theta = \frac{a_1 b_1 + a_2 b_2 + a_3 b_3}{\sqrt{a_1^2 + a_2^2 + a_3^2} \sqrt{b_1^2 + b_2^2 + b_3^2}}$$

➤ Let $(\vec{a}, \vec{b}) = \theta$, then

$$(i) \theta < 90^\circ \Leftrightarrow \vec{a} \cdot \vec{b} > 0 \quad (ii) \theta > 90^\circ \Leftrightarrow \vec{a} \cdot \vec{b} < 0$$

$$(iii) \theta = 90^\circ \Leftrightarrow \vec{a} \cdot \vec{b} = 0 \quad (\vec{a} \text{ is perpendicular to } \vec{b})$$

W.E-1: If $\vec{a} = \vec{i} - \lambda \vec{j} + 2\vec{k}$ and

$\vec{b} = 8\vec{i} + 6\vec{j} - \vec{k}$ are perpendicular then λ is

Sol: Let $\vec{a} = \vec{i} - \lambda \vec{j} + 2\vec{k}$ and

$\vec{b} = 8\vec{i} + 6\vec{j} - \vec{k}$ Since, $\vec{a}, \vec{b}$ are at right angles, $\vec{a} \cdot \vec{b} = 0$

$$\Rightarrow (\vec{i} - \lambda \vec{j} + 2\vec{k}) \cdot (8\vec{i} + 6\vec{j} - \vec{k}) = 0$$

$$\Rightarrow 8 - 6\lambda - 2 = 0 \Rightarrow \lambda = 1$$

➤ If $\vec{a}$ and $\vec{b}$ are parallel and are in the same direction ($\vec{a}, \vec{b}$ are like vectors) then $\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}|$

➤ If $\vec{a}$ and $\vec{b}$ are parallel and are in the opposite direction ($\vec{a}, \vec{b}$ are unlike vectors) then $\vec{a} \cdot \vec{b} = -|\vec{a}| |\vec{b}|$

➤ Let $\vec{a}, \vec{b}$ are two unit vectors and $\vec{a} \cdot \vec{b} = \cos \theta$

➤ If $\vec{a}, \vec{b}$ are any two vectors, then $\vec{a} \cdot \vec{b} = 0$ either $\vec{a} = \vec{0}$ (or) $\vec{b} = \vec{0}$ (or) $\vec{a} \perp \vec{b}$. However if $\vec{a}, \vec{b}$ are two non-zero perpendicular vectors, then $\vec{a} \cdot \vec{b} = 0$

➤ **Geometrical interpretation of scalar product :**

Let $\vec{a}$ and $\vec{b}$ be two vectors represented by $\vec{OA}$ and $\vec{OB}$ respectively. Let θ be the angle between $\vec{OA}$ and $\vec{OB}$. Draw

$\vec{BL} \perp \vec{OA}$ and $\vec{AM} \perp \vec{OB}$. From triangle OBL and OAM, We have $OL = OB \cos \theta$ and

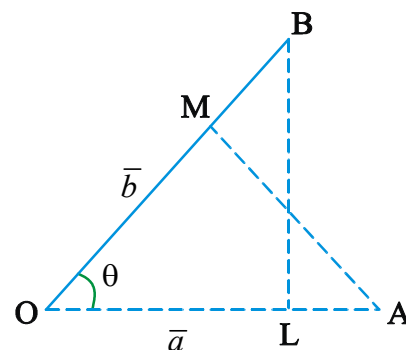
$OM = OA \cos \theta$. Here OL and OM are known as projections of $\vec{b}$ on $\vec{a}$ and $\vec{a}$ on $\vec{b}$.

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta = |\vec{a}| (OB \cos \theta) = |\vec{a}| (OL) \\ = (\text{magnitude of } \vec{a}) (\text{projection of } \vec{b} \text{ on } \vec{a})$$

$$\text{again } \vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta = |\vec{b}| (|\vec{a}| \cos \theta)$$

$$= |\vec{b}| (OA \cos \theta) = |\vec{b}| (OM)$$

$$= (\text{magnitude of } \vec{b}) (\text{projection of } \vec{a} \text{ on } \vec{b})$$



➤ Let $\vec{a}$ and $\vec{b}$ be two nonzero vectors. Then Component vector of $\vec{b}$ on $\vec{a}$ (or) orthogonal

$$\text{projection of } \vec{b} \text{ on } \vec{a} \text{ is } \frac{(\vec{b} \cdot \vec{a}) \vec{a}}{|\vec{a}|^2}$$

WE-2: Orthogonal projection of

$\vec{b} = 2\vec{i} - 3\vec{j} + 6\vec{k}$ on $\vec{a} = \vec{i} + 2\vec{j} + 2\vec{k}$ is

Sol: Orthogonal projection

$$= \frac{[(2\vec{i} - 3\vec{j} + 6\vec{k}) \cdot (\vec{i} + 2\vec{j} + 2\vec{k})](\vec{i} + 2\vec{j} + 2\vec{k})}{|\vec{i} + 2\vec{j} + 2\vec{k}|^2}$$

$$= \frac{(2 - 6 + 12)}{9}(\vec{i} + 2\vec{j} + 2\vec{k}) = \frac{8}{9}(\vec{i} + 2\vec{j} + 2\vec{k})$$

➤ Component vector of $\vec{a}$ on $\vec{b}$ (or) orthogonal

projection of $\vec{a}$ on $\vec{b}$ is $\frac{(\vec{a} \cdot \vec{b})\vec{b}}{|\vec{b}|^2}$

WE-3: If $\vec{a} = 2\vec{i} + \vec{j} + 2\vec{k}$, $\vec{b} = 5\vec{i} - 3\vec{j} + \vec{k}$

then orthogonal projection of $\vec{a}$ on $\vec{b}$ is

Sol: Orthogonal projection of $\vec{a}$ on $\vec{b}$ is $\frac{(\vec{a} \cdot \vec{b})\vec{b}}{|\vec{b}|^2} =$

$$= \frac{(10 - 3 + 2)(5\vec{i} - 3\vec{j} + \vec{k})}{25 + 9 + 1} = \frac{9(5\vec{i} - 3\vec{j} + \vec{k})}{35}$$

➤ The orthogonal projection of $\vec{b}$ in the direction

perpendicular to that of $\vec{a}$ is $\vec{b} - \frac{(\vec{b} \cdot \vec{a})\vec{a}}{|\vec{a}|^2}$

WE-4: The orthogonal projection of

$\vec{b} = 3\vec{i} + 2\vec{j} - 5\vec{k}$ on a vector perpendicular to

$\vec{a} = 2\vec{i} - \vec{j} + 2\vec{k}$ is

Sol: Orthogonal projection of $\vec{b}$ on $\vec{a}$

$$= 3\vec{i} + 2\vec{j} - 5\vec{k} - \frac{(3\vec{i} + 2\vec{j} - 5\vec{k}) \cdot (2\vec{i} - \vec{j} + 2\vec{k})}{4 + 1 + 4} (2\vec{i} - \vec{j} + 2\vec{k})$$

$$= 3\vec{i} + 2\vec{j} - 5\vec{k} - \frac{6 - 2 - 10}{9} (2\vec{i} - \vec{j} + 2\vec{k})$$

$$= (3\vec{i} + 2\vec{j} - 5\vec{k}) + \frac{2}{3}(2\vec{i} - \vec{j} + 2\vec{k})$$

$$= \frac{13\vec{i} + 4\vec{j} - 11\vec{k}}{3}$$

➤ The length of the orthogonal projection of $\vec{b}$ on $\vec{a}$

is $\left| \frac{(\vec{a} \cdot \vec{b})}{|\vec{a}|} \right|$

➤ The length of the orthogonal projection of $\vec{a}$ on $\vec{b}$

is $\left| \frac{(\vec{a} \cdot \vec{b})}{|\vec{b}|} \right|$

WE-5: The length of orthogonal projection of

$\vec{a} = 2\vec{i} - 3\vec{j} + \vec{k}$ on $\vec{b} = 4\vec{i} - 4\vec{j} + 7\vec{k}$ is

Sol: The length of the orthogonal projection of $\vec{a}$

on $\vec{b}$ is $\left| \frac{(\vec{a} \cdot \vec{b})}{|\vec{b}|} \right| = \frac{|8 + 12 + 7|}{\sqrt{16 + 16 + 49}} = \frac{27}{9} = 3$

➤ The scalar product is commutative i.e., $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$

➤ The scalar product is distributive over vector addition i.e., $\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$, $(\vec{b} + \vec{c}) \cdot \vec{a} = \vec{b} \cdot \vec{a} + \vec{c} \cdot \vec{a}$

➤ $(l\vec{a}) \cdot \vec{b} = \vec{a} \cdot (l\vec{b}) = l(\vec{a} \cdot \vec{b})$ where l is a scalar

➤ $\vec{a} \cdot \vec{a} \geq 0$; $|\vec{a} \cdot \vec{b}| \leq |\vec{a}| |\vec{b}|$; $|\vec{a} - \vec{b}| \geq ||\vec{a}| - |\vec{b}||$

$$|\vec{a} + \vec{b}| \leq |\vec{a}| + |\vec{b}|$$

➤ Cauchy schwartz in equality : Let

a_1, a_2, a_3 and b_1, b_2, b_3 be real numbers. Then

$$(a_1 b_1 + a_2 b_2 + a_3 b_3)^2 \leq$$

$$(a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2) \text{ and equality}$$

holds If $\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3}$

➤ $|\vec{a} + \vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 + 2\vec{a} \cdot \vec{b}$

➤ $|\vec{a} - \vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 - 2\vec{a} \cdot \vec{b}$

➤ $|\vec{a} + \vec{b} + \vec{c}|^2 = |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a})$

➤ Let l_1, m_1, n_1 be the direction cosines of $\vec{a}$ and let l_2, m_2, n_2 be the direction cosines of $\vec{b}$ and let $(\vec{a}, \vec{b}) = \theta$ then $\cos \theta = l_1 l_2 + m_1 m_2 + n_1 n_2$

➤ The vector equation to the plane which is at a distance of p units from the origin and $\hat{n}$ is a unit vector perpendicular to the plane is $\vec{r} \cdot \hat{n} = p$

➤ If the origin lies on the plane then its equation is $\vec{r} \cdot \vec{n} = 0$

- The vector equation of a plane passing through the point $A(\vec{a})$ and perpendicular to the vector $\vec{n}$ is $(\vec{r} - \vec{a}) \cdot \vec{n} = 0$

W.E-6 : The vector equation of the plane passing through the point $(3, -2, 1)$ and perpendicular to the vector $(4, 7, -4)$ is ____

Sol. $\{\vec{r} - (3\vec{i} - 2\vec{j} + \vec{k})\} \cdot (4\vec{i} + 7\vec{j} - 4\vec{k}) = 0$
 $\Rightarrow \vec{r} \cdot (4\vec{i} + 7\vec{j} - 4\vec{k}) =$
 $(3\vec{i} - 2\vec{j} + \vec{k}) \cdot (4\vec{i} + 7\vec{j} - 4\vec{k}) = 12 - 14 - 4$
 $\Rightarrow \vec{r} \cdot (4\vec{i} + 7\vec{j} - 4\vec{k}) = -6$

- In a parallelogram, if its diagonals are equal then it is a rectangle.
- In a parallelogram, the sum of the squares of the diagonals is equal to the sum of the squares of the sides.
- If $\vec{F}$ be the force and $\vec{s}$ be the displacement inclined at an angle θ with the direction of the force, then work done $\vec{F} \cdot \vec{s}$
- If a constant force $\vec{F}$ acting on a particle displaces it from A to B, then work done, $W = \vec{F} \cdot \vec{AB}$
- If $\vec{F}$ is the resultant of the forces $\vec{F}_1, \vec{F}_2, \dots, \vec{F}_n$ then work done in displacing the particle from A to B is $W = (\vec{F}_1 + \vec{F}_2 + \dots + \vec{F}_n) \cdot \vec{AB}$
- If a number of forces are acting on a particle, the sum of the work done by the separate forces is equal to the work done by the resultant force.
- A line makes angles $\alpha, \beta, \gamma, \delta$, with the four diagonals of a cube then $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + \cos^2 \delta = 4/3$
- If $\vec{r}$ is any vector then $\vec{r} = (\vec{r} \cdot \vec{i})\vec{i} + (\vec{r} \cdot \vec{j})\vec{j} + (\vec{r} \cdot \vec{k})\vec{k}$
- If $\vec{a}, \vec{b}$ are two vectors then
- i) $\vec{a} \cdot \vec{a} \geq 0$ ii) $|\vec{a} \cdot \vec{b}| \leq |\vec{a}| |\vec{b}|$
- iii) $|\vec{a} + \vec{b}| \leq |\vec{a}| + |\vec{b}|$ iv) $|\vec{a} - \vec{b}| \geq ||\vec{a}| - |\vec{b}||$
- The cartesian equation of the plane passing through

the point $A(x_1, y_1, z_1)$ and perpendicular to the vector $\vec{m} = a\vec{i} + b\vec{j} + c\vec{k}$ is

$$a(x - x_1) + b(y - y_1) + c(z - z_1) = 0.$$

- The equation of the plane passing through the point $A(x_1, y_1, z_1)$ and whose normal has d.r.s a, b, c is $a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$.
- Angle between any two diagonals of a cube is $\cos^{-1}(1/3)$.
- Angle between a diagonal of a cube and a diagonal of a face of the cube which are passes through the same corner is $\cos^{-1} \sqrt{2/3}$.
- Angle between a diagonal of a cube and edge of a cube is $\cos^{-1} \left(\frac{1}{\sqrt{3}} \right)$

Angle between a line and a plane :

- i) The angle between a line and a plane is the complement of the angle between the line and normal to the plane. If θ is the angle between a line $\vec{r} = \vec{a} + t\vec{b}$ and a plane $\vec{r} \cdot \vec{m} = d$ then $\cos(90^\circ - \theta) = \sin \theta = \frac{|\vec{b} \cdot \vec{m}|}{|\vec{b}| |\vec{m}|}$.

WE-7: The angle between the line

$$\vec{r} = (-i + 3j + 3k) + t(2i + 3j + 6k)$$

and the plane $\vec{r} \cdot (-\vec{i} + \vec{j} + \vec{k}) = 5$ is

Sol: $\sin \theta = \frac{(2\vec{i} + 3\vec{j} + 6\vec{k}) \cdot (-\vec{i} + \vec{j} + \vec{k})}{|2\vec{i} + 3\vec{j} + 6\vec{k}| |-\vec{i} + \vec{j} + \vec{k}|} =$
 $\frac{-2 + 3 + 6}{\sqrt{4 + 9 + 36} \sqrt{1 + 1 + 1}} = \frac{1}{\sqrt{3}} \Rightarrow \theta = \sin^{-1} \left(\frac{1}{\sqrt{3}} \right)$

- If θ is the angle between the line $\frac{x - x_1}{l} = \frac{y - y_1}{m} = \frac{z - z_1}{n}$ and the plane $ax + by + cz + d = 0$

then i) $\sin \theta = \frac{al + bm + cn}{\sqrt{a^2 + b^2 + c^2} \sqrt{l^2 + m^2 + n^2}}$

ii) If the line is perpendicular to the plane then

$$\frac{l}{a} = \frac{m}{b} = \frac{n}{c}.$$

iii) If the line is parallel to the plane then

$$al + bm + cn = 0$$

➤ The perpendicular distance of the plane $\vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n}$ from the origin is $\frac{|\vec{a} \cdot \vec{n}|}{|\vec{n}|}$

➤ Angle between the planes $\vec{r} \cdot \vec{n}_1 = p_1, \vec{r} \cdot \vec{n}_2 = p_2$, then $\cos \theta = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1| |\vec{n}_2|}$

WE-8: Angle between the planes

$\vec{r} \cdot (2\vec{i} - \vec{j} + \vec{k}) = 3$ and $\vec{r} \cdot (\vec{i} + \vec{j} + 2\vec{k}) = 4$ is

Sol: Let $\vec{a} = 2\vec{i} - \vec{j} + \vec{k}$ and

$$\vec{b} = \vec{i} + \vec{j} + 2\vec{k}$$

If θ is the angle between the planes then

$$\cos \theta = \cos(\vec{a}, \vec{b}) = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} =$$

$$\frac{2-1+2}{\sqrt{4+1+1}\sqrt{1+1+4}} = \frac{3}{6} = \frac{1}{2} \Rightarrow \theta = 60^\circ$$



C.U.Q

1. If $|\vec{a} + \vec{b}| < |\vec{a} - \vec{b}|$ then the angle between the vectors $\vec{a}, \vec{b}$ is

- 1) Acute Angle 2) Obtuse Angle
3) Right Angle 4) 45°

2. $R(\vec{r})$ is any point on the semi-circle $P(\vec{p})$

and $Q(\vec{q})$ are the position vectors of diameter

of that semicircle. Then $\overline{PR} \cdot \overline{QR}$ is equal to

- 1) 1 2) 0 3) 3 4) Not defined

3. The non-zero vectors $\vec{a}, \vec{b}, \vec{c}$ are related by $\vec{a} = 8\vec{b}$ and $\vec{c} = -7\vec{b}$, then the angle between $\vec{a}$ & $\vec{c}$ is (AIE-2008)

- 1) 0 2) $\frac{\pi}{4}$ 3) $\frac{\pi}{2}$ 4) π

4. If $(\vec{a}, \vec{b}) = 0^\circ$ or 180° then $\vec{a}, \vec{b}$ are

- 1) Perpendicular 2) Parallel
3) Parallel and are in the same direction
4) Parallel and are in the opposite direction

5. If $\vec{a} \cdot \vec{b} = -|\vec{a}| |\vec{b}|$ then the vectors $\vec{a}$ and $\vec{b}$ are

- 1) Like vectors 2) Unlike vectors
3) Equal vectors 4) Perpendicular vectors

6. If $\vec{a}, \vec{b}, \vec{c}$ are three non-zero vectors then $\vec{a} \cdot \vec{b} = \vec{a} \cdot \vec{c}$ implies that

- 1) $\vec{a}$ is orthogonal to $(\vec{b} + \vec{c})$
2) $\vec{a}$ is orthogonal to both $\vec{b}$ and $\vec{c}$
3) $\vec{b} = \vec{c} + \vec{a}$
4) $\vec{a}$ is orthogonal to $(\vec{b} - \vec{c})$ (or) $\vec{b} = \vec{c}$

7. Equality holds in the triangle inequality

$$|\vec{a} + \vec{b}| \leq |\vec{a}| + |\vec{b}| \text{ if}$$

- 1) $\vec{a} = m\vec{b}$ 2) $\vec{a} = m\vec{b}, m > 0$
3) $\vec{a} = m\vec{b}, m < 0$ 4) $(\vec{a}, \vec{b}) = 90^\circ$

8. The vector equation $\vec{a} \cdot \vec{b} = \vec{a} \cdot \vec{c}$ need not always imply

- 1) $\vec{a} = 0$ 2) $\vec{b} = \vec{c}$
3) $\vec{a} \perp (\vec{b} - \vec{c})$ 4) All the above

9. Let $\vec{a}$ and $\vec{b}$ be two nonzero vectors. Then $(\vec{a} \cdot \vec{b})^2$ is

- 1) $\geq |\vec{a}|^2 |\vec{b}|^2$ 2) $= |\vec{a}|^2 |\vec{b}|^2$ 3) $\leq |\vec{a}|^2 |\vec{b}|^2$ 4) $= |\vec{a}|^2 |\vec{b}|^2$

10. If $\vec{a}, \vec{b}, \vec{c}$ are mutually perpendicular and $\vec{r}$

is any vector, then $\sum \left(\frac{\vec{r} \cdot \vec{a}}{\vec{a} \cdot \vec{a}} \right) \vec{a} = \dots$

- 1) $\vec{r}$ 2) $\vec{a}$ 3) $\vec{b}$ 4) $\vec{c}$

11. The equation of the plane passing through the point with position vector $\vec{a}$ and perpendicular to $\vec{b}$ is (EAM-1990)

- 1) $\vec{r} \cdot (\vec{a} \times \vec{b}) = 0$ 2) $\vec{r} = \vec{a} \times \vec{b}$
3) $\vec{r} = \vec{b} \times \vec{a}$ 4) $(\vec{r} - \vec{a}) \cdot \vec{b} = 0$

12. The angle between the straight lines

$\vec{r} = \vec{a} + t\vec{u}, \vec{r} = \vec{a} + s\vec{v}$ is θ then

- 1) $\sin \theta = \frac{|\vec{u} \cdot \vec{v}|}{|\vec{u}| |\vec{v}|}$ 2) $\cos \theta = \frac{|\vec{u} \cdot \vec{v}|}{|\vec{u}| |\vec{v}|}$
3) $\cos \theta = \vec{u} \cdot \vec{v}$ 4) $\sin \theta = \vec{u} \cdot \vec{v}$

13. If $|\vec{a}| + |\vec{b}| = |\vec{c}|$ and $\vec{a} + \vec{b} = \vec{c}$ then the angle between $\vec{a}$ and $\vec{b}$

- 1) 0° 2) $\frac{\pi}{6}$ 3) $\frac{\pi}{3}$ 4) $\frac{\pi}{2}$

C.U.Q-KEY

- 01) 2 02) 2 03) 4 04) 2 05) 2 06) 4
07) 2 08) 4 09) 3 10) 1 11) 4 12) 2
13) 1

C.U.Q-HINTS

- squaring and expand
- Angle in a semicircle is a right angle.
- since $\vec{a}$ & $\vec{c}$ have opposite direction $(\vec{a}, \vec{c}) = \pi$.
- Definition of parallel vectors
- $\cos \theta = -1 \Rightarrow \theta = 180^\circ$
- $\vec{a} \cdot \vec{b} - \vec{a} \cdot \vec{c} = 0 \Rightarrow \vec{a} \cdot (\vec{b} - \vec{c}) = 0$
- $(a, b) = 0 \Rightarrow$ vector inequality concept
 $a = mb, m > 0$
- $\vec{a} \cdot \vec{b} = \vec{a} \cdot \vec{c}$ then (1) $\vec{a} = 0$ (2) $\vec{b} = \vec{c}$ (3) $\vec{a} \perp \vec{b}, \vec{c}$
the three results may be true or may not be true.
- $|\vec{a} \cdot \vec{b}| \leq |\vec{a}| |\vec{b}| \Rightarrow (a \cdot b)^2 \leq |a|^2 |b|^2$
- $\vec{r} = \sum (\vec{r} \cdot \hat{a}) \hat{a}$
- Vector equation of plane passing through $\vec{a}$ and perpendicular $\vec{b}$ is $(\vec{r} - \vec{a}) \cdot \vec{b} = 0$ or $\vec{r} \cdot \vec{b} = \vec{a} \cdot \vec{b}$
- $\cos \theta = \frac{\vec{u} \cdot \vec{v}}{|\vec{u}| |\vec{v}|}$ of the lines passing through $\vec{a}$ and parallel to $\vec{u}, \vec{v}$.
- $\vec{a} + \vec{b} = \vec{c}$ Squaring on both sides



LEVEL - I (C.W)

SCALAR (OR) DOT PRODUCT OF VECTORS GEOMETRICAL INTERPRETATION
ORTHOGONAL PROJECTION

- If $\vec{a} = \vec{i} + 2\vec{j} + \vec{k}$, $\vec{b} = 2\vec{j} + \vec{k} - \vec{i}$, then component of $\vec{a}$ perpendicular to $\vec{b}$ is
1) $\frac{5}{2}\vec{i} + \frac{2}{3}\vec{j} + \frac{1}{2}\vec{k}$ 2) $\frac{5}{3}\vec{i} + 2\vec{j} + \frac{1}{3}\vec{k}$
3) $\frac{5\vec{i} + 2\vec{j} + \vec{k}}{3}$ 4) $\vec{i} + \vec{j} + \vec{k}$
- The orthogonal projection of $\vec{a} = 2\vec{i} + 3\vec{j} + 3\vec{k}$ on $\vec{b} = \vec{i} - 2\vec{j} + \vec{k}$ (where $\vec{i}, \vec{j}, \vec{k}$ are unit vectors along three mutually perpendicular directions) is (EAM-1996)

$$1) \frac{-\vec{i} + 2\vec{j} - \vec{k}}{6} \quad 2) \frac{-\vec{i} + 2\vec{j} - \vec{k}}{\sqrt{6}}$$

$$3) \vec{i} - 2\vec{j} + \vec{k} \quad 4) -\vec{i} + 2\vec{j} - \vec{k}$$

- Given two vectors $\vec{a} = 2\vec{i} - 3\vec{j} + 6\vec{k}$, $\vec{b} = -2\vec{i} + 2\vec{j} - \vec{k}$ and $\lambda = \frac{\text{the projection of } \vec{a} \text{ on } \vec{b}}{\text{the projection of } \vec{b} \text{ on } \vec{a}}$, then the value of λ is

$$1) \frac{3}{7} \quad 2) 7 \quad 3) 3 \quad 4) \frac{7}{3}$$

- If the vector OP in XY plane whose magnitude is $\sqrt{3}$ makes an angle 60° with Y-axis, the length of the component of the vector in direction of X-axis is

$$1) 1 \quad 2) \sqrt{3} \quad 3) 1/2 \quad 4) 3/2$$

- If $\vec{a} = 4\vec{i} + 6\vec{j}$ and $\vec{b} = 3\vec{j} + 4\vec{k}$, then the vector form of the component of a along b is

$$1) \frac{18(3\vec{i} + 4\vec{k})}{10\sqrt{3}} \quad 2) \frac{18(3\vec{j} + 4\vec{k})}{25}$$

$$3) \frac{18(3\vec{i} + 4\vec{k})}{\sqrt{13}} \quad 4) \frac{(3\vec{j} + 4\vec{k})}{25}$$

- If $\vec{p} = (2, 1, 3)$, $\vec{q} = (-2, 3, 1)$, $\vec{r} = (3, -2, 4)$ and $\vec{j}$ is the unit vector in the direction of y-axis, then $(2\vec{p} + 3\vec{q} - 4\vec{r}) \cdot \vec{j} =$

$$1) 18 \quad 2) 19 \quad 3) 20 \quad 4) 21$$

- The projection of the vector $\vec{a} = 4\vec{i} - 3\vec{j} + 2\vec{k}$ on the vector making equal angles (acute) with coordinate axes and having magnitude $\sqrt{3}$ is

$$1) 3 \quad 2) \sqrt{3} \quad 3) 2\sqrt{3} \quad 4) 1$$

- The sum of the length of projections of $p\vec{i} + q\vec{j} + r\vec{k}$ on the coordinate axes, where $p = 2, q = 3$ and $r = 1$ is

$$1) 6 \quad 2) 5 \quad 3) 4 \quad 4) 3$$

- Let P = (1, 0, -1) Q = (-1, 2, 0) R = (2, 0, -3) S = (3, -2, -1), then the length of the component of RS on PQ is

$$1) 1/3 \quad 2) 2/3 \quad 3) 4/3 \quad 4) 5/3$$

10. The vectors $\vec{a}, \vec{b}, \vec{c}$ are of the same length and taken pair wise, they form equal angles. If $\vec{a} = \vec{i} + \vec{j}$ and $\vec{b} = \vec{j} + \vec{k}$ then the components of $\vec{c}$ are

- 1) (1, 0, 1) 2) (1, 2, 3)
3) (-1, 1, 2) 4) (-1, 4, 1)

PROPERTIES OF DOT PRODUCT

11. If $\vec{a}$ is collinear with $\vec{b} = 3\vec{i} + 6\vec{j} + 6\vec{k}$ and $\vec{a} \cdot \vec{b} = 27$. Then $\vec{a}$ is equal to

- 1) $3(\vec{i} + \vec{j} + \vec{k})$ 2) $\vec{i} + 3\vec{j} + 3\vec{k}$
3) $\vec{i} + 2\vec{j} + 2\vec{k}$ 4) $2\vec{i} + 2\vec{j} + 2\vec{k}$

12. Let $\vec{a}$ and $\vec{b}$ be two unit vectors and θ be the angle between them then $|\vec{a} + \vec{b}|^2 - |\vec{a} - \vec{b}|^2 =$

- 1) $\cos \theta$ 2) $2 \cos \theta$ 3) $3 \cos \theta$ 4) $4 \cos \theta$

13. The vector $\vec{x}$ which is perpendicular to (2,-3,1) and (1,-2,3) and which satisfies the condition $\vec{x} \cdot (\vec{i} + 2\vec{j} - 7\vec{k}) = 10$

- 1) $3\vec{i} + 5\vec{j} + \vec{k}$ 2) $7\vec{i} - 5\vec{j} + \vec{k}$
3) $3\vec{i} - 5\vec{j} + \vec{k}$ 4) $7\vec{i} + 5\vec{j} + \vec{k}$

14. $A = 4\vec{i} + 5\vec{j} + 6\vec{k}$, $B = 5\vec{i} + 6\vec{j} + 4\vec{k}$
 $C = 6\vec{i} + 4\vec{j} + 5\vec{k}$ are the vertices of

- 1) Scalene triangle 2) Equilateral triangle
3) Right angled triangle 4) Isosceles triangle

15. If θ is acute angle and the vector $(\sin \theta)\vec{i} + (\cos \theta)\vec{j}$ is perpendicular to the vector $\vec{i} - \sqrt{3}\vec{j}$ then $\theta =$ (EAM-2000)

- 1) $\frac{\pi}{6}$ 2) $\frac{\pi}{5}$ 3) $\frac{\pi}{4}$ 4) $\frac{\pi}{3}$

16. $\vec{a}, \vec{b}, \vec{c}$ are three vectors, such that $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, $|\vec{a}| = 1$, $|\vec{b}| = 2$, $|\vec{c}| = 3$, then

- $(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a})^2$ is equal to
1) -7 2) 49 3) 7 4) 1

17. If $\vec{A} \cdot (\vec{B} + \vec{C}) = \vec{B} \cdot (\vec{C} + \vec{A}) = \vec{C} \cdot (\vec{A} + \vec{B}) = 0$ and $|\vec{A}| = 3$, $|\vec{B}| = 4$ and $|\vec{C}| = 5$ then $|\vec{A} + \vec{B} + \vec{C}| =$
1) 5 2) $5\sqrt{2}$ 3) $5/\sqrt{2}$ 4) $\sqrt{2}$

18. If $|\vec{a}| = 3$, $|\vec{b}| = 4$ and $|\vec{a} - \vec{b}| = 5$ then $|\vec{a} + \vec{b}| =$ (EAM 1994)
1) 6 2) 5 3) 4 4) 3

19. If $|\vec{a}| = 1$, $|\vec{b}| = 2$, $|\vec{a} - \vec{b}|^2 + |\vec{a} + 2\vec{b}|^2 = 20$, then $(\vec{a}, \vec{b}) =$

- 1) $\frac{\pi}{3}$ 2) $\frac{\pi}{4}$ 3) $\frac{\pi}{6}$ 4) $\frac{2\pi}{3}$

20. Let $|\vec{a}| = 3$ and $|\vec{b}| = 4$. The value of ' μ ' for which the vectors $\vec{a} + \mu\vec{b}$ and $\vec{a} - \mu\vec{b}$ are perpendicular is...

- 1) $\frac{3}{4}$ 2) $\frac{2}{3}$ 3) $\pm \frac{3}{4}$ 4) $-\frac{2}{3}$

21. If $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ are the vertices of a square then

- 1) $\vec{b} - \vec{a} = \vec{c} - \vec{b}$ 2) $\vec{a} + \vec{b} = \vec{c}$
3) $(\vec{c} - \vec{a}) \cdot (\vec{d} - \vec{b}) = 0$ 4) $\vec{a} + \vec{b} = 0$

22. If $\vec{a} + \vec{b}$ is perpendicular to $\vec{b}$ and $\vec{a} + 2\vec{b}$ is perpendicular to $\vec{a}$ then.

- 1) $|\vec{a}| = |\vec{b}|$ 2) $|\vec{a}| = \sqrt{2}|\vec{b}|$
3) $|\vec{b}| = \sqrt{2}|\vec{a}|$ 4) $|\vec{a}| = |\vec{b}|\sqrt{3}$

23. $(\vec{a} - \vec{d}) \cdot (\vec{b} - \vec{c}) + (\vec{b} - \vec{d}) \cdot (\vec{c} - \vec{a}) + (\vec{c} - \vec{d}) \cdot (\vec{a} - \vec{b}) =$

- 1) $\vec{0}$ (null vector) 2) 0
3) $\vec{a} \cdot \vec{b} + \vec{c} \cdot \vec{d}$ 4) $\vec{a} \cdot \vec{c} + \vec{b} \cdot \vec{d}$

24. If $\vec{a}, \vec{b}, \vec{c}$ are unit vectors, then

$|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2$ does not exceed

- 1) 4 2) 9 3) 8 4) 6

25. Dot product of a vector with vectors $3\vec{i} - 5\vec{k}$, $2\vec{i} + 7\vec{j}$ and $\vec{i} + \vec{j} + \vec{k}$ are respectively -1, 6 and 5. The vector is

- 1) $3\vec{i} + \vec{j} + 2\vec{k}$ 2) $3\vec{i} + 2\vec{k}$
3) $3\vec{i} - 2\vec{k}$ 4) $\vec{i} + \vec{j} + \vec{k}$

26. The vector $\vec{b}$ which is collinear with the vector $\vec{a} = (2, 1, -1)$ and satisfies the relation $\vec{a} \cdot \vec{b} = 3$ is ...

- 1) $\left(1, \frac{1}{2}, \frac{-1}{2}\right)$ 2) $\left(\frac{2}{3}, \frac{1}{3}, \frac{-1}{3}\right)$
3) $\left(\frac{1}{2}, \frac{1}{4}, \frac{-1}{4}\right)$ 4) (1, 1, 0)

27. If two adjacent sides of a square are represented by the vectors $x\vec{i} + \vec{j} + 4\vec{k}$ and $3\vec{i} + y\vec{j}$ then $xy =$

- 1) 1 2) 2 3) 3 4) -3

28. ABCD is a rhombus. If $\vec{AC} = \vec{i} + (1 + \lambda)\vec{j} + (\lambda - 2)\vec{k}$ and $\vec{BD} = (2\lambda - 1)\vec{i} + \vec{j} - \vec{k}$, then $\lambda =$

- 1) 1 2) -1 3) 2 4) -2

29. If the sum of two unit vectors is a unit vector, then the magnitude of their difference is

- 1) 3 2) $\sqrt{3}$ 3) $\sqrt{13}$ 4) $\sqrt{7}$

SCALAR (DOT) PRODUCT ANGLE BETWEEN TWO VECTORS:

30. If $|\vec{a}| = |\vec{b}| = |\vec{a} - \vec{b}|$, then the angle between $\vec{a}$ and $\vec{b}$ is

- 1) $\frac{\pi}{4}$ 2) $\frac{\pi}{2}$ 3) $\frac{\pi}{3}$ 4) $\frac{2\pi}{3}$

31. If $\vec{a}$ and $\vec{b}$ are two unit vectors and θ be the angle between them, then $\sin\left(\frac{\theta}{2}\right) =$

- 1) $|\vec{a} - \vec{b}|$ 2) $|\vec{a} + \vec{b}|$ 3) $\frac{1}{2}|\vec{a} - \vec{b}|$ 4) $\frac{1}{2}|\vec{a} + \vec{b}|$

32. If $\vec{a}$ and $\vec{b}$ are unit vectors and θ is the angle between them, then $\vec{a} - \vec{b}$ will be a unit vector if $\theta =$ (EAM-1997)

- 1) $\frac{\pi}{4}$ 2) $\frac{\pi}{3}$ 3) $\frac{\pi}{6}$ 4) $\frac{\pi}{2}$

33. If $\vec{a} = p\vec{i} + 3\vec{j} - 7\vec{k}$, $\vec{b} = p\vec{i} - p\vec{j} + 4\vec{k}$ and if the angle between $\vec{a}$ and $\vec{b}$ is acute, then the values of p lies in

- 1) $P < -4$ or $p > 7$ 2) $(-7, 4)$
3) $P \leq -4$ or $p \geq 7$ 4) $[-7, 4]$

34. If $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, $|\vec{a}| = 3$, $|\vec{b}| = 5$ and $|\vec{c}| = 7$ then the angle between $\vec{a}$ and $\vec{b}$ is

- 1) 30° 2) 45° 3) 60° 4) 90°

35. If $\vec{a}, \vec{b}, \vec{c}$ are three unit vectors such that $|\vec{a} + \vec{b} + \vec{c}| = 1$ and $\vec{a} \perp \vec{b}$. If $\vec{c}$ makes angles α, β with $\vec{a}, \vec{b}$ respectively, then $\cos \alpha + \cos \beta$ is equal to

- 1) $\frac{3}{2}$ 2) 1 3) -1 4) $-\frac{3}{2}$

36. If $\vec{AB} = (3, -2, 2)$, $\vec{BC} = (-1, 0, -2)$ are the adjacent sides of a parallelogram, then the obtuse angle between its diagonals is

- 1) $\frac{\pi}{4}$ 2) $\pi/2$ 3) $\pi/3$ 4) $\frac{3\pi}{4}$

37. If $\vec{a}, \vec{b}, \vec{c}$ are three mutually perpendicular vectors such that $|\vec{a}| = |\vec{b}| = |\vec{c}|$ then $(\vec{a} + \vec{b} + \vec{c}, \vec{a}) =$

- 1) $\frac{\pi}{3}$ 2) $\cos^{-1}\left(\frac{1}{3}\right)$
3) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$ 4) $\cos^{-1}\left(\frac{2}{3}\right)$

38. If $\vec{a}$ and $\vec{b}$ are unit vectors inclined to x-axis at angle 30° and 120° , then $|\vec{a} + \vec{b}|$ equals

- 1) $\sqrt{2/3}$ 2) $\sqrt{2}$ 3) $\sqrt{3}$ 4) 2

39. If $\vec{AB} = -\vec{i} - 2\vec{j} - 6\vec{k}$, $\vec{BC} = 2\vec{i} - \vec{j} + \vec{k}$, $\vec{AC} = \vec{i} - 3\vec{j} - 5\vec{k}$. Then $\angle B =$

- 1) $\cos^{-1}\left(\sqrt{\frac{40}{41}}\right)$ 2) $\cos^{-1}\left(\sqrt{\frac{6}{41}}\right)$
3) $\cos^{-1}\left(\frac{6}{41}\right)$ 4) $\cos^{-1}\left(\frac{62}{63}\right)$

40. If $\vec{v}$ and $\vec{w}$ are two mutually perpendicular unit vectors and $\vec{u} = a\vec{v} + b\vec{w}$, where a and b are non zero real numbers, then the angle between $\vec{u}$ and $\vec{w}$ is

- 1) $\cos^{-1}\left(\frac{b}{\sqrt{a^2 + b^2}}\right)$ 2) $\cos^{-1}\left(\frac{a}{\sqrt{a^2 + b^2}}\right)$
3) $\cos^{-1}(b)$ 4) $\cos^{-1}(a)$

41. The value of a , for which the points A, B, C with position vectors $2\vec{i} - \vec{j} + \vec{k}$, $\vec{i} - 3\vec{j} - 5\vec{k}$ and $a\vec{i} - 3\vec{j} + \vec{k}$ respectively are the vertices of a right angled triangle with $\angle C = \frac{\pi}{2}$ are

- 1) -2 or 1 2) 2 or -1
3) 2 or 1 4) -2 or -1

(AIE-2006)

42. The points O, A, B, C, D are such that $\overrightarrow{OA} = \vec{a}$, $\overrightarrow{OB} = \vec{b}$, $\overrightarrow{OC} = 2\vec{a} + 3\vec{b}$ and $\overrightarrow{OD} = \vec{a} - 2\vec{b}$. If $|\vec{a}| = 3|\vec{b}|$, then the angle between $\overrightarrow{BD}$ and $\overrightarrow{AC}$ is

- 1) π 2) $\frac{\pi}{2}$ 3) $\frac{\pi}{3}$ 4) $\frac{\pi}{6}$

43. If $\vec{a} = (-1, 1, 2)$; $\vec{b} = (2, 1, -1)$; $\vec{c} = (-2, 1, 3)$ then the angle between $2\vec{a} - \vec{c}$ and $\vec{a} + \vec{b}$ is

- 1) $\frac{\pi}{4}$ 2) $\frac{\pi}{3}$ 3) $\frac{\pi}{6}$ 4) $\frac{3\pi}{2}$

VECTOR EQUATION OF A PLANE NORMAL FORM

44. The cartesian equation of the plane perpendicular to vector $3\vec{i} - 2\vec{j} - 2\vec{k}$ and passing through the point $2\vec{i} + 3\vec{j} - \vec{k}$ is

- 1) $3x + 2y + 2z = 2$ 2) $3x - 2y + 2z = 2$
3) $3x + 2y - 2z = 2$ 4) $3x - 2y - 2z = 2$

45. The perpendicular distance from origin to the plane $3x - 2y - 2z = 2$ is

- 1) $1/\sqrt{17}$ 2) $2/\sqrt{17}$ 3) $3/\sqrt{17}$ 4) $4/\sqrt{17}$

46. The vector equation of the plane which is perpendicular to $2\vec{i} - 3\vec{j} + \vec{k}$ and at a distance of 5 units from the origin is (EAM-1991)

- 1) $\vec{r} \cdot (2\vec{i} - 3\vec{j} + \vec{k}) = 5\sqrt{14}$ 2) $\vec{r} \cdot (2\vec{i} - 3\vec{j} + \vec{k}) = 5$
3) $\vec{r} \cdot \frac{(2\vec{i} - 3\vec{j} + \vec{k})}{\sqrt{14}}$ 4) $\vec{r} \cdot \frac{(2\vec{i} + 3\vec{j} + \vec{k})}{\sqrt{14}}$

ANGLE BETWEEN THE PLANES

47. The angle between planes

$\vec{r} \cdot (2\vec{i} - 3\vec{j} + 4\vec{k}) + 11 = 0$ and $\vec{r} \cdot (3\vec{i} - 2\vec{j} - 3\vec{k}) + 27 = 0$ is

- 1) $\pi/6$ 2) $\pi/3$ 3) $\pi/4$ 4) $\pi/2$

APPLICATIONS OF SCALAR PRODUCT IN MECHANICS TO FIND THE WORK DONE:

48. A particle is acted upon by constant forces $4\vec{i} + \vec{j} - 3\vec{k}$ and $3\vec{i} + \vec{j} - \vec{k}$ which displace it from a point $\vec{i} + 2\vec{j} + 3\vec{k}$ to the point $5\vec{i} + 4\vec{j} + \vec{k}$. The work done in standard units by the forces is given by (AIE-2004)

- 1) 40 2) 30 3) 25 4) 15

LOCUS

49. The locus of the point equidistant from two given points a and b is given by

1) $[\vec{r} - 1/2(\vec{a} + \vec{b})] \cdot (\vec{a} - \vec{b}) = 0$

2) $[\vec{r} - 1/2(\vec{a} - \vec{b})] \cdot (\vec{a} + \vec{b}) = 0$

3) $[\vec{r} - 1/2(\vec{a} + \vec{b})] \cdot (\vec{a} + \vec{b}) = 0$

4) $[\vec{r} - 1/2(\vec{a} - \vec{b})] \cdot (\vec{a} - \vec{b}) = 0$

50. If the vectors $\vec{i} - 2\vec{x}\vec{j} - 3\vec{y}\vec{k}$ and $\vec{i} + 3\vec{x}\vec{j} - 2\vec{y}\vec{k}$ are orthogonal to each other, then the locus of the point (x, y) is

- 1) Circle 2) Ellipse
3) Hyperbola 4) Pair of lines

LEVEL-I (C.W)-KEY

- 01) 3 02) 1 03) 4 04) 4 05) 2 06) 2
07) 2 08) 1 09) 3 10) 1 11) 3 12) 4
13) 4 14) 2 15) 4 16) 2 17) 2 18) 2
19) 4 20) 3 21) 3 22) 2 23) 2 24) 2
25) 2 26) 1 27) 4 28) 2 29) 2 30) 3
31) 3 32) 2 33) 1 34) 3 35) 3 36) 4
37) 3 38) 2 39) 2 40) 1 41) 3 42) 2
43) 3 44) 4 45) 2 46) 1 47) 4 48) 1
49) 1 50) 3

LEVEL-I (C.W)-HINTS

1. $\vec{a} - \frac{(\vec{a} \cdot \vec{b})\vec{b}}{|\vec{b}|^2}$

2. Orthogonal projection of $\vec{a}$ on $\vec{b} = \frac{(\vec{a} \cdot \vec{b})\vec{b}}{|\vec{b}|^2}$

3. $\lambda = \frac{\vec{a} \cdot \vec{b}}{\vec{b} \cdot \vec{a}} = \frac{|\vec{a}|}{|\vec{b}|}$

4. $\sqrt{3} \cos(90^\circ - 60^\circ)$

5. $\frac{(\vec{a} \cdot \vec{b})\vec{b}}{|\vec{b}|^2}$

6. $2\vec{p} + 3\vec{q} - 4\vec{r} = -14\vec{i} + 19\vec{j} - 7\vec{k}$

7. $\frac{\sqrt{3}(\vec{i} + \vec{j} + \vec{k}) \cdot \vec{a}}{|\vec{i} + \vec{j} + \vec{k}|} = \sqrt{3}$

8. Use $p + q + r$
9. $\frac{|\overline{RS} \cdot \overline{PQ}|}{|\overline{PQ}|}$
10. $|\vec{a}| = \sqrt{2}, |\vec{b}| = \sqrt{2}, |(1, 0, 1)| = \sqrt{1+0+1} = \sqrt{2}$
11. $\vec{a} = x\vec{i} + y\vec{j} + z\vec{k}$ and $\vec{a} \cdot \vec{b} = 27$.
12. $|\vec{a} + \vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 + 2\vec{a} \cdot \vec{b}$
13. $\vec{x} = x\vec{i} + y\vec{j} + z\vec{k}$ and $\vec{x} \cdot (\vec{i} + 2\vec{j} - 7\vec{k}) = 10$
14. find $|\vec{a}|, |\vec{b}|, |\vec{c}|$
15. $(\sin \theta)(1) + \cos \theta(-\sqrt{3}) = 0$
 $\Rightarrow \frac{\sin \theta}{\cos \theta} = \sqrt{3} \Rightarrow \theta = 60^\circ$
16. $|\vec{a} + \vec{b} + \vec{c}| = 0 \Rightarrow 1 + 4 + 9 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) = 0$
 $\Rightarrow \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = -7$
17. $2(\vec{A} \cdot \vec{B} + \vec{B} \cdot \vec{C} + \vec{C} \cdot \vec{A}) = 0$
 $|\vec{A} + \vec{B} + \vec{C}|^2 = 9 + 16 + 25$
18. $|\vec{a} + \vec{b}|^2 + |\vec{a} - \vec{b}|^2 = 2(|\vec{a}|^2 + |\vec{b}|^2)$
19. On expanding $\vec{a} \cdot \vec{b} = -1, \cos(\vec{a}, \vec{b}) = -\frac{1}{2}$
 $(\vec{a}, \vec{b}) = \frac{2\pi}{3}$
20. Use $|\vec{a}|^2 = |\mu^2| |\vec{b}|^2$
21. Diagonals are Perpendicular $\overline{AC} \cdot \overline{BD} = 0$
22. $(\vec{a} + \vec{b}) \cdot \vec{b} = 0$ and $(\vec{a} + 2\vec{b}) \cdot \vec{a} = 0$ simplify
23. $\vec{a} \cdot \vec{b} - \vec{a} \cdot \vec{c} - \vec{d} \cdot \vec{b} + \vec{d} \cdot \vec{c} + \vec{b} \cdot \vec{c} - \vec{b} \cdot \vec{a} - \vec{d} \cdot \vec{c} + \vec{d} \cdot \vec{a} + \vec{c} \cdot \vec{a} - \vec{c} \cdot \vec{b} - \vec{d} \cdot \vec{a} + \vec{d} \cdot \vec{b} = 0$
24. $|\vec{a} + \vec{b} + \vec{c}|^2 \geq 0 \Rightarrow \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} \geq -\frac{3}{2}$
 Now, $|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2 \leq 9$
25. Check the dot product from options.
26. Check from options that $\vec{b} = \lambda \vec{a}$ and $\vec{a} \cdot \vec{b} = 3$
27. dot product = 0

28. $\overline{AC} \cdot \overline{BD} = 0$
29. use $|\vec{a} + \vec{b}|^2 + |\vec{a} - \vec{b}|^2 = 2(|\vec{a}|^2 + |\vec{b}|^2)$
30. $|\vec{a} - \vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 - 2|\vec{a}||\vec{b}|\cos \theta$
 $\Rightarrow |\vec{a}|^2 = |\vec{a}|^2 + |\vec{b}|^2 - 2|\vec{a}||\vec{b}|\cos \theta$
 $+ |\vec{b}|^2 = +2|\vec{a}||\vec{b}|\cos \theta, \cos \theta = \frac{1}{2} \Rightarrow \theta = 60^\circ$
31. $\sin \frac{\theta}{2} = \frac{1}{2} |\vec{a} - \vec{b}|$
32. $|\vec{a} - \vec{b}| = 1 \Rightarrow |\vec{a} - \vec{b}|^2 = 1$
33. $\vec{a} \cdot \vec{b} > 0$
34. $\vec{a} + \vec{b} + \vec{c} = \vec{0} \Rightarrow \vec{a} + \vec{b} = -\vec{c}$
 S.O.B.S and expand
35. $|\vec{a} + \vec{b} + \vec{c}|^2 = |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a})$
36. Find $\overline{AC}$ & $\overline{BD} \therefore \cos \theta = \frac{\overline{AC} \cdot \overline{BD}}{|\overline{AC}||\overline{BD}|}$
37. $\cos(\vec{a} + \vec{b} + \vec{c}, \vec{a}) = \frac{(\vec{a} + \vec{b} + \vec{c}) \cdot \vec{a}}{|\vec{a} + \vec{b} + \vec{c}||\vec{a}|}$
 $= \frac{|\vec{a}|^2}{\sqrt{3}|\vec{a}|^2} = \frac{1}{\sqrt{3}}$
38. Clearly, $\vec{a}$ and $\vec{b}$ are at right angles
 $\therefore |\vec{a} + \vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 + 2|\vec{a}||\vec{b}|\cos 90^\circ$
 $\Rightarrow |\vec{a} + \vec{b}|^2 = 1 + 1 + 0 = 2 \Rightarrow |\vec{a} + \vec{b}| = \sqrt{2}$
39. $\cos \theta = \frac{\overline{BC} \cdot \overline{BA}}{|\overline{BC}||\overline{BA}|}$
40. $\vec{u} \cdot \vec{w} = a(\vec{v} \cdot \vec{w}) + b(\vec{w} \cdot \vec{w}) = b$
 $\Rightarrow \cos \theta = \frac{b}{|\vec{u}||\vec{w}|} = \frac{b}{\sqrt{a^2 + b^2}}$
41. $\overline{CA} \cdot \overline{CB} = 0$
42. Use $\cos \theta = \frac{\overline{BD} \cdot \overline{AC}}{|\overline{BD}||\overline{AC}|}$
43. $2\vec{a} - \vec{c} = (0, 1, 1), \vec{a} + \vec{b} = (1, 2, 1)$
 then $\cos \theta = \frac{2+1}{\sqrt{2}\sqrt{6}} \Rightarrow \cos^{-1}(\sqrt{3}/2), \theta = 30^\circ$

44. Use $(\vec{r} - \vec{a}) \cdot \vec{b} = 0$

45. Distance = $\frac{|d|}{\sqrt{a^2 + b^2 + c^2}}$

46. $\vec{a} = 2\vec{i} - 3\vec{j} + \vec{k}$; $p = 5 \Rightarrow$ plane equation

$$\vec{r} \cdot \frac{\vec{a}}{|\vec{a}|} = p$$

47. $\frac{\vec{m}_1 \cdot \vec{m}_2}{|\vec{m}_1||\vec{m}_2|} = \cos \theta$

48. $\vec{F} = \vec{F}_1 + \vec{F}_2$ and W.E. = $\vec{F} \cdot \vec{d}$

49. The locus of a point which is equidistant from the given two points is the perpendicular bisector of the line segment joining the two points i.e., $\vec{MP} \cdot \vec{BA} = 0$ ($\because M$ is mid point)

50. $\vec{a} \cdot \vec{b} = 0$

$6x^2 - 6y^2 = 1$, hyperbola



LEVEL - I (H.W)

SCALAR (OR) DOT PRODUCT OF VECTORS GEOMETRICAL INTERPRETATION ORTHOGONAL PROJECTION:

1. If $\vec{a} = 2\vec{i} + \vec{j} + 2\vec{k}$ and $\vec{b} = 5\vec{i} - 3\vec{j} + \vec{k}$, then the projection of $\vec{b}/2$ on $\vec{a}$ is

- 1) 1 2) 2 3) $3/2$ 4) 4

2. If $\vec{a} = 2\vec{i} + 3\vec{j}$ and $\vec{b} = 3\vec{j} + 4\vec{k}$. Then the vector form of the component of $\vec{a}$ along $\vec{b}$ is

- 1) $\frac{9(3\vec{j} + 4\vec{k})}{10\sqrt{3}}$ 2) $\frac{9(3\vec{j} + 4\vec{k})}{25}$
 3) $\frac{9(3\vec{j} + 4\vec{k})}{\sqrt{13}}$ 4) $3\vec{j} + 4\vec{k}$

3. If $\vec{a} = 2\vec{i} + \vec{j} + \vec{k}$, $\vec{b} = \vec{i} + 5\vec{j}$, $\vec{c} = 4\vec{i} + 4\vec{j} - 2\vec{k}$ then the length of the projection of

$(3\vec{a} - 2\vec{b})$ in the direction of $\vec{c}$

- 1) 3 2) -3 3) 33 4) -33

4. The angle between $\vec{a}$ and $\vec{b}$ is $\frac{5\pi}{6}$ and the projection of $\vec{a}$ on $\vec{b}$ is $\frac{-6}{\sqrt{3}}$ then $|\vec{a}| =$

- 1) 6 2) $\frac{\sqrt{3}}{2}$ 3) 12 4) 4

5. The component of $a\vec{i} + b\vec{j} + c\vec{k}$ on the Y-axis is

- 1) a 2) b 3) c 4) 0

6. The components of a vector on the co-ordinate axes are 2, 1, 2. Then the length of the vector is

- (1) 3 (2) 1 (3) 2 (4) 5

7. A vector has 3 units length and equal components then the vector is

- (1) $\vec{i} + \vec{j} + \vec{k}$ (2) $\sqrt{2}(\vec{i} + \vec{j} + \vec{k})$
 (3) $\sqrt{3}(\vec{i} + \vec{j} + \vec{k})$ (4) $\sqrt{5}(\vec{i} + \vec{j} + \vec{k})$

8. The length of the orthogonal projection of a vector $\vec{i} - 2\vec{j} + \vec{k}$ on vector $4\vec{i} - 4\vec{j} + 7\vec{k}$ is

- 1) $\frac{19}{6}$ 2) $\frac{19}{7}$
 3) $\frac{19(4\vec{i} - 4\vec{j} + 7\vec{k})}{9}$ 4) $\frac{19}{9}$

9. Let $\vec{a} = \vec{i} - \vec{j} + 3\vec{k}$ and $\vec{b} = 3\vec{i} - 5\vec{j} + 6\vec{k}$. Then the magnitude of the projection of $2\vec{a} - \vec{b}$ on $\vec{a} + \vec{b}$ is

- 1) $\frac{22}{\sqrt{133}}$ 2) $\frac{11\sqrt{2}}{\sqrt{10}}$ 3) $\frac{22}{\sqrt{10}}$ 4) $\frac{22}{\sqrt{5}}$

10. $\vec{a}, \vec{b}, \vec{c}$ are three vectors such that $|\vec{a}| = 1$, $|\vec{b}| = 2$, $|\vec{c}| = 3$, and $\vec{b}, \vec{c}$ are perpendicular to each other. If the projection of $\vec{b}$ along $\vec{a}$ is same as that of $\vec{c}$ along $\vec{a}$ then $|\vec{a} - \vec{b} + \vec{c}|$ is equal to

- 1) $\sqrt{2}$ 2) $\sqrt{7}$ 3) $\sqrt{14}$ 4) 14

11. The vector component of $\vec{a} = 2\vec{i} + 3\vec{j} + 2\vec{k}$ perpendicular to the direction of $\vec{b} = \vec{i} + 2\vec{j} + \vec{k}$ is

- 1) $\frac{1}{3}(\vec{i} + \vec{j} + \vec{k})$ 2) $\frac{1}{3}(\vec{i} - \vec{j} + \vec{k})$
 3) $\frac{1}{3}(\vec{i} + \vec{j} - \vec{k})$ 4) $\frac{1}{3}(-\vec{i} + \vec{j} - \vec{k})$

12. If $\vec{a} = 2\vec{i} + \vec{j} + \vec{k}$, $\vec{b} = \vec{i} + 5\vec{j}$,
 $\vec{c} = 4\vec{i} + 4\vec{j} - 2\vec{k}$ then the length of the
 projection of $(3\vec{a} - 2\vec{b})$ in the direction of $\vec{c}$
 (1) 3 (2) -3 (3) 33 (4) -33

PROPERTIES OF DOT PRODUCT

13. In $\triangle ABC$, $A = \vec{a}$, $B = \vec{b}$, $C = \vec{c}$. If $P(\vec{r})$ is any
 point in the plane of $\triangle ABC$ such that
 $\vec{b} \cdot \vec{c} + \vec{a} \cdot \vec{r} = \vec{c} \cdot \vec{a} + \vec{b} \cdot \vec{r} = \vec{a} \cdot \vec{b} + \vec{c} \cdot \vec{r}$ then P is of
 $\triangle ABC$.
 1) In-Center 2) Circum Center
 3) Ortho Center 4) Centroid
14. If two out of the 3 vectors $\vec{a}, \vec{b}, \vec{c}$ are unit
 vectors, $\vec{a} + \vec{b} + \vec{c} = \vec{0}$ and
 $2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) + 3 = 0$ then the length of the
 third vector is
 1) 3 2) 2 3) 1 4) 0
15. If $\vec{a}, \vec{b}, \vec{c}$ are unit vectors such that
 $\vec{a} + \vec{b} + \vec{c} = \vec{0}$ then $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} =$
 1) 3/2 2) -3/2 3) $2[\vec{a} \cdot \vec{b} \cdot \vec{c}]$ 4) 0
16. If the scalar product of the vector $\vec{i} + \vec{j} + \vec{k}$ with
 the unit vector parallel to the sum of the
 vectors $2\vec{i} + 4\vec{j} - 5\vec{k}$ and $x\vec{i} + 2\vec{j} + 3\vec{k}$ is equal to 1
 then x =
 1) 0 2) 1 3) 2 4) 3
17. If $\vec{r} = (x+y+2)\vec{i} + (2x-y+3)\vec{j} + (x+2y+7)\vec{k}$
 where $\vec{r} \cdot \vec{i} = 3$, $\vec{r} \cdot \vec{j} = 5$ then $\vec{r} \cdot \vec{k} =$
 1) 4 2) 6 3) 9 4) 8
18. If ABCD is a parallelogram and
 $\overline{AC}^2 - \overline{BD}^2 = K \overline{AB}$. (Orthogonal projection
 of $\overline{AD}$ on $\overline{AB}$) then K =
 1) 1 2) 2 3) 3 4) 4
19. If the vectors $a\vec{i} - 2\vec{j} + \vec{k}$, $2a\vec{i} + a\vec{j} - 4\vec{k}$ are
 adjacent sides of a rectangle, then the values
 of 'a' are
 1) 1, 2 2) -1, 2 3) 1, -2 4) 2, -2
20. If $|\vec{a} + \vec{b}| = 1$, $|\vec{a} - \vec{b}| = 7$, $|\vec{a}| = 4$, then $|\vec{b}| =$
 1) 1 2) 2 3) 3 4) 5

21. If $\vec{a} \cdot \vec{i} = \vec{a} \cdot (\vec{i} + \vec{j}) = \vec{a} \cdot (\vec{i} + \vec{j} + \vec{k})$ then $\vec{a} =$ (EAM-2002)
 1) $\vec{i}$ 2) $\vec{j}$ 3) $\vec{k}$ 4) $\vec{i} + \vec{j} + \vec{k}$
22. The vector $\vec{b}$ which is collinear with the vector
 $\vec{a} = (2, -1, 2)$ and satisfies the relation $\vec{a} \cdot \vec{b} = 18$ is
 1) $(4, -2, 4)$ 2) $(2, 1, -1)$
 3) $(1, -1, 1)$ 4) $(1, 1, 0)$
23. If $\vec{a}, \vec{b}, \vec{c}$ are orthogonal and $\vec{r} \cdot \vec{a} = 2$ then
 $\vec{r} - (\vec{r} \cdot \vec{b})\vec{b} - (\vec{r} \cdot \vec{c})\vec{c} =$
 1) $2\vec{a}$ 2) $-2\vec{a}$ 3) $\vec{a}$ 4) $-\vec{a}$

SCALAR (DOT) PRODUCT ANGLE BETWEEN TWO VECTORS:

24. If $\vec{a}$ and $\vec{b}$ are two unit vectors inclined at an
 angle θ to each other, then $|\vec{a} + \vec{b}| < 1$ if
 1) $\theta = \frac{\pi}{6}$ 2) $\theta = \frac{\pi}{2}$
 3) $\theta = \frac{\pi}{3}$ 4) $\frac{2\pi}{3} < \theta \leq \pi$
25. Let $\vec{e}_1, \vec{e}_2$ be unit vectors which include angle
 θ . If $\frac{1}{2}|\vec{e}_1 - \vec{e}_2| = \sin(k\theta)$, then k equal to
 1) 2 2) 1 3) $\frac{1}{2}$ 4) $\frac{1}{\sqrt{2}}$
26. If $\vec{a}$ and $\vec{b}$ are two vectors of lengths 2 and 1
 respectively and $|\vec{a} - \vec{b}| = \sqrt{3}$ then $(\vec{a}, \vec{b}) =$
 1) $\frac{\pi}{6}$ 2) $\frac{\pi}{4}$ 3) $\frac{\pi}{3}$ 4) $\frac{\pi}{2}$
27. The angle between $8(\vec{i} - \vec{k})$ and $\vec{i} + \vec{j} + \vec{k}$ is
 1) $\frac{\pi}{4}$ 2) $\frac{\pi}{3}$ 3) $\frac{\pi}{2}$ 4) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$
28. Angle between the vectors
 $\vec{a} = -\vec{i} - 2\vec{j} + \vec{k}$ and $\vec{b} = x\vec{i} + \vec{j} + (x+1)\vec{k}$
 1) Obtuse angle 2) Acute angle
 3) Right angle 4) Depends on x

29. If $\vec{a}$ and $\vec{b}$ are non-zero and different vectors such that $|\vec{a} + \vec{b}| = |\vec{b} - \vec{a}|$ then the angle between $-\vec{a}$ and $\vec{b}$ is
- 1) $\frac{\pi}{3}$ 2) $\frac{\pi}{4}$ 3) $\frac{\pi}{2}$ 4) $\frac{\pi}{6}$
30. The acute angles made by the line joining the points $(1, -3, 2)$ and $(3, -5, 1)$ with the coordinate axes are
- 1) $\cos^{-1}\left(\frac{2}{3}\right), \cos^{-1}\left(\frac{2}{3}\right), \cos^{-1}\left(\frac{1}{3}\right)$
 2) $\cos^{-1}\left(\frac{3}{2}\right), \cos^{-1}\left(\frac{2}{3}\right), \cos^{-1}\left(\frac{2}{3}\right)$
 3) $\cos^{-1}\left(\frac{1}{3}\right), \cos^{-1}\left(\frac{3}{2}\right), \cos^{-1}\left(\frac{1}{3}\right)$
 4) $\cos^{-1}\left(\frac{2}{3}\right), \cos^{-1}\left(\frac{3}{2}\right), \cos^{-1}\left(\frac{1}{3}\right)$
31. If the angle between the vectors $(x, 3, -7)$ and $(x, -x, 4)$ is obtuse, the domain of 'x' is
- 1) $(-4, 7)$ 2) $[-4, 7]$ 3) $R - [-4, 7]$ 4) $R - (-4, 7)$
32. If $\vec{a}, \vec{b}, \vec{c}$ are three vectors such that $\vec{a} = \vec{b} + \vec{c}$ and the angle between $\vec{b}$ and $\vec{c}$ is $\frac{\pi}{2}$, then (here $a = |\vec{a}|, b = |\vec{b}|, c = |\vec{c}|$)
- 1) $a^2 = b^2 + c^2$ 2) $b^2 = c^2 + a^2$
 3) $c^2 = a^2 + b^2$ 4) $2a^2 - b^2 = c^2$
33. If the position vectors of A, B and C are respectively $2\vec{i} - \vec{j} + \vec{k}, \vec{i} - 3\vec{j} - 5\vec{k}$ and $3\vec{i} - 4\vec{j} - 4\vec{k}$ then $\cos^2 A =$
- 1) 0 2) $\frac{6}{41}$ 3) $\frac{35}{41}$ 4) 1
34. If $\vec{a} = \vec{i} + 2\vec{j} - 3\vec{k}$ and $\vec{b} = 3\vec{i} - \vec{j} + 2\vec{k}$ then the angle between the vectors $(\vec{a} + \vec{b})$ and $(\vec{a} - \vec{b})$ is
- 1) $\frac{\pi}{4}$ 2) $\frac{\pi}{3}$ 3) $\frac{\pi}{2}$ 4) $\frac{\pi}{6}$
35. If the angle between $\vec{a}$ and $\vec{b}$ is 120° $|\vec{a}| = 3$ and $|\vec{b}| = 5$, then $|\vec{a} - \vec{b}|$ is equals to
- 1) 6 2) 7 3) 9 4) 5
36. If $\vec{a} = 2\vec{i} + 2\vec{j} - \vec{k}$ and $\vec{b} = 6\vec{i} - 3\vec{j} + 2\vec{k}$ then the angle between $\vec{a}$ and $\vec{b}$ is
- 1) $\cos^{-1}\frac{1}{3}$ 2) $\frac{\pi}{3}$
 3) $\cos^{-1}\left(\frac{4}{21}\right)$ 4) $\cos^{-1}\left(\frac{5\sqrt{7}}{21}\right)$
37. If $\vec{e}_1$ and $\vec{e}_2$ are unit vectors and the vectors $\vec{e}_1 + 2\vec{e}_2, 5\vec{e}_1 - 4\vec{e}_2$ are at right angles, then the angle between $\vec{e}_1$ and $\vec{e}_2$ is
- 1) 30° 2) 60° 3) 45° 4) 75°
38. If $\vec{a}, \vec{b}, \vec{c}$ are unit vectors satisfying the relation $\vec{a} + \vec{b} + \sqrt{3}\vec{c} = 0$, then the angle between $\vec{a}$ and $\vec{b}$ is
- 1) $\frac{\pi}{6}$ 2) $\frac{\pi}{4}$ 3) $\frac{\pi}{3}$ 4) $\frac{\pi}{2}$
39. The angle between the vectors $\vec{a} = 6\vec{i} + 2\vec{j} + 3\vec{k}$ and $\vec{b} = 2\vec{i} - 9\vec{j} + 6\vec{k}$ is
- 1) $\cos^{-1}\left(\frac{2}{3}\right)$ 2) $\cos^{-1}\left(\frac{7}{12}\right)$
 3) $\cos^{-1}\left(\frac{12}{77}\right)$ 4) $\cos^{-1}\left(\frac{1}{3}\right)$
40. Angle between $\vec{a}$ and $\vec{b}$ is 120° . If $|\vec{b}| = 2|\vec{a}|$ and the vectors $\vec{a} + x\vec{b}, \vec{a} - \vec{b}$ are at right angles, then x is equal to
- 1) $\frac{1}{3}$ 2) $\frac{1}{5}$ 3) $\frac{2}{5}$ 4) $\frac{2}{3}$
41. Let $A(2, -3, 0) B(2, -1, 1) C(0, 1, 4)$ be the vertices of $\triangle ABC$. Then angle between the median $\overline{BD}$ and the base $\overline{AC}$ is
- (1) 30° (2) 45° (3) 60° (4) 90°

VECTOR EQUATION OF A PLANE NORMAL FORM

42. The cartesian equation of the plane passing through the point $(3, -2, 1)$ and perpendicular to vector $4\vec{i} + 7\vec{j} - 4\vec{k}$ is

1) $4x + 7y + 4z - 6 = 0$ 2) $4x + 7y - 4z + 6 = 0$
3) $4x - 7y - 4z - 6 = 0$ 4) $4x - 7y + 4z + 6 = 0$

43. The perpendicular distance from origin to the plane $3x - 4y - 5z = 3$ is

1) $\frac{3}{5}$ 2) 3 3) $\frac{3}{4\sqrt{2}}$ 4) $\frac{3}{5\sqrt{2}}$

44. The perpendicular distance from origin to the plane $3x + 4y + 5z = 25$ is

(1) $\frac{3}{5}$ (2) 3 (3) $\frac{3}{4\sqrt{2}}$ (4) $\frac{5}{\sqrt{2}}$

ANGLE BETWEEN THE PLANES

45. Angle between the planes $2x - y + 2z = 3$, $3x + 6y + 2z = 4$ is

1) $\cos^{-1}\left(\frac{4}{21}\right)$ 2) $\cos^{-1}\left(\frac{4}{441}\right)$

3) $\sin^{-1}\left(\frac{4}{21}\right)$ 4) $\sin^{-1}\left(\frac{4}{441}\right)$

46. Angle between the plane $x - y + z = 3$, $x + y + z = 4$ is

(1) $\cos^{-1} \frac{1}{3}$ (2) $\cos^{-1} \frac{4}{41}$

(3) $\sin^{-1} \frac{4}{27}$ (4) $\sin^{-1} \frac{4}{41}$

47. If the vectors $\vec{i} - 2x\vec{j} + 3y\vec{k}$ and

$\vec{i} - 2x\vec{j} - 3y\vec{k}$ are perpendicular then the

locus of (x, y) is :

- (1) A circle (2) An ellipses
(3) A parabola (4) A hyperbola

APPLICATIONS OF SCALAR PRODUCT IN MECHANICS TO FIND THE WORK DONE:

48. The force $\vec{F} = 3\vec{i} + \vec{j} - \vec{k}$ acts on a particle and it moves from the point $A(2\vec{i} - \vec{j})$ to

$B(2\vec{i} + \vec{j})$. The work done by the force $\vec{F} =$

1) 1 2) 2 3) 3 4) 4

49. The work done by force $\vec{F} = a\vec{i} + \vec{j} + \vec{k}$ in moving a particle from $(1, 1, 1)$ to $(2, 2, 2)$ along a straight line is 5 units. Then $a =$
(1) 1 (2) 2 (3) 3 (4) 4

LEVEL - I (H.W)-KEY

01) 3	02) 2	03) 1	04) 4	05) 2	06) 1
07) 3	08) 4	09) 1	10) 3	11) 2	12) 1
13) 3	14) 3	15) 2	16) 2	17) 4	18) 4
19) 2	20) 3	21) 1	22) 1	23) 1	24) 4
25) 3	26) 3	27) 3	28) 1	29) 3	30) 1
31) 1	32) 1	33) 3	34) 3	35) 2	36) 3
37) 2	38) 3	39) 3	40) 3	41) 2	42) 2
43) 4	44) 4	45) 1	46) 1	47) 4	48) 2
49) 3					

LEVEL - I (H.W)-HINTS

- $\frac{\vec{b} \cdot \vec{a}}{|\vec{a}|}$
- $\frac{(\vec{a} \cdot \vec{b})\vec{b}}{|\vec{b}|^2}$
- $\frac{|(3\vec{a} - 2\vec{b}) \cdot \vec{c}|}{|\vec{c}|} = 3$
- $\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} = \frac{-6}{\sqrt{3}} \Rightarrow \frac{|\vec{a}||\vec{b}|\left(-\frac{\sqrt{3}}{2}\right)}{|\vec{b}|} = \frac{-6}{\sqrt{3}} \Rightarrow |\vec{a}| = 4$
- $\frac{(\vec{a}\vec{i} + \vec{b}\vec{j} + \vec{c}\vec{k}) \cdot \vec{j}}{|\vec{j}|} = b$
- $\sqrt{a^2 + b^2 + c^2} = \sqrt{4 + 1 + 4} = 3$
- $\frac{3(\vec{i} + \vec{j} + \vec{k})}{\sqrt{3}}$
- $\frac{|\vec{a} \cdot \vec{b}|}{|\vec{b}|}$
- $\vec{a} + \vec{b} = 4\vec{i} - 6\vec{j} + 9\vec{k}, 2\vec{a} - \vec{b} = \vec{i} + 3\vec{j}$
 $(2\vec{a} - \vec{b}) \cdot (\vec{a} + \vec{b}) = -4 - 18 = -22$
 $= \frac{|(2\vec{a} - \vec{b}) \cdot (\vec{a} + \vec{b})|}{|\vec{a} + \vec{b}|} = \frac{22}{\sqrt{133}}$

10. The projection of $\vec{b}$ along $\vec{a} = \left(\frac{(\vec{a} \cdot \vec{b})}{|\vec{a}|^2} \right) \vec{a}$
11. Component of $\vec{a}$ perpendicular to the direction of $\vec{b}$

$$\vec{a} = \vec{a} - \frac{(\vec{a} \cdot \vec{b})}{|\vec{b}|^2} \vec{b}$$
12. $\frac{|(3\vec{a} - 2\vec{b}) \cdot \vec{c}|}{|\vec{c}|} = 3$
13. Here $\vec{r} \cdot (\vec{a} - \vec{b}) = \vec{c} \cdot (\vec{a} - \vec{b}) \Rightarrow \vec{r} - \vec{c}$
 perpendicular to $\vec{a} - \vec{b} \Rightarrow \overline{CP}$ Perpendicular to $\overline{AB}$. Similarly $\overline{AP}$ Perpendicular to $\overline{AB} \Rightarrow \overline{CP}$ and $\overline{AP}$ are altitudes $\therefore 'P' =$
 Ortho Center of $\triangle ABC$.
14. $|\vec{a} + \vec{b} + \vec{c}|^2 = 0 \Rightarrow 1 + 1 + |\vec{c}|^2 - 3 = 0 \Rightarrow |\vec{c}| = 1$
15. $|\vec{a} + \vec{b} + \vec{c}|^2 = 0 \Rightarrow |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) = 0$
16. $(\vec{i} + \vec{j} + \vec{k}) \cdot \frac{((2+x)\vec{i} + 6\vec{j} - 2\vec{k})}{|(2+x)\vec{i} + 6\vec{j} - 2\vec{k}|} = 1$
17. $\left. \begin{array}{l} \vec{r} \cdot \vec{i} = 3 \dots (1) \\ \vec{r} \cdot \vec{j} = 5 \dots (2) \end{array} \right\} \text{Solving (1) \& (2).}$
18. Use orthogonal projection formula.
19. Dot product = 0.
20. $|\vec{a} + \vec{b}|^2 + |\vec{a} - \vec{b}|^2 = 2(|\vec{a}|^2 + |\vec{b}|^2)$
 $\Rightarrow 1 + 49 = 2(16 + |\vec{b}|^2) \Rightarrow 2|\vec{b}|^2 = 18 \Rightarrow |\vec{b}| = 3.$
21. Let $\vec{a} = x\vec{i} + y\vec{j} + z\vec{k}$ then $x = 1, y = 0, z = 0$
 so $\vec{a} = \vec{i}$
22. Check from options that $\vec{b} = \lambda \vec{a}$ and $\vec{a} \cdot \vec{b} = 3.$
23. Use $\vec{r} = (\vec{r} \cdot \vec{a})\vec{a} + (\vec{r} \cdot \vec{b})\vec{b} + (\vec{r} \cdot \vec{c})\vec{c}.$
24. $|\vec{a} + \vec{b}| < 1$ squaring on both sides.
25. Conceptual
26. Expand.
27. $\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}$

28. $\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}$
29. Here $\vec{a}$ perpendicular $\vec{b} \Rightarrow (\vec{a}, \vec{b}) = \frac{\pi}{2}$
 $\Rightarrow (-\vec{a}, \vec{b}) = \pi - \frac{\pi}{2} = \frac{\pi}{2}$
30. Find $\cos(\overline{AB}, \vec{i}), \cos(\overline{AB}, \vec{j}), \cos(\overline{AB}, \vec{k}).$
31. $(x, 3, -7) \cdot (x, -x, 4) < 0.$
32. $|\vec{a}|^2 = |\vec{b} + \vec{c}|^2 \Rightarrow a^2 = b^2 + c^2 + 2\vec{b} \cdot \vec{c}$
 $a^2 = b^2 + c^2 [\because \vec{b} \cdot \vec{c} = 0].$
33. $\cos A = \frac{\overline{AB} \cdot \overline{AC}}{|\overline{AB}| |\overline{AC}|}.$
34. Check $(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) = 0$ (or) check $|\vec{a}| = |\vec{b}|.$
36. $|\vec{a} - \vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 - 2(\vec{a} \cdot \vec{b}) = 34 + 15 = 49$
 Therefore, $|\vec{a} - \vec{b}| = 7$
36. $\vec{a} \cdot \vec{b} = 12 - 6 - 2 = 4$ $|\vec{a}| = 3, |\vec{b}| = 7$
 $\therefore \cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} = \frac{4}{21}$
37. $(\vec{e}_1 + 2\vec{e}_2) \cdot (5\vec{e}_1 - 4\vec{e}_2) = 0 \Rightarrow 5 - 8 + 6 \cos \theta$
 where, $\theta = (\vec{e}_1, \vec{e}_2) \Rightarrow \cos \theta = \frac{1}{2} \Rightarrow \theta = 60^\circ$
38. $\vec{a} + \vec{b} = -\sqrt{3}\vec{c} \Rightarrow (\vec{a} + \vec{b})^2 = 3c^2$
 $\Rightarrow 1 + 1 + 2(\vec{a} \cdot \vec{b}) = 3 \Rightarrow \cos \theta = \frac{1}{2} \Rightarrow \theta = 60^\circ$
39. $\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}$
40. $\vec{a} \cdot \vec{b} = 2|\vec{a}|^2 \cos 120^\circ = -|\vec{a}|^2$
 $(\vec{a} - x\vec{b}) \cdot (\vec{a} - \vec{b}) = 0 \Rightarrow x = \frac{2}{5}$
41. $\theta = \cos^{-1} \left(\frac{\overline{BD} \cdot \overline{AC}}{|\overline{BD}| |\overline{AC}|} \right)$

42. $(\bar{r} - \bar{a}) \cdot \bar{b} = 0$

$$\Rightarrow [(x-3)\bar{i} + (y+2)\bar{j} + (z-1)\bar{k}] \cdot (4\bar{i} + 7\bar{j} - 4\bar{k}) = 0.$$

43. Distance = $\frac{|d|}{\sqrt{a^2 + b^2 + c^2}}$

44. Distance = $\frac{|d|}{\sqrt{a^2 + b^2 + c^2}}$

45. $\cos \theta = \frac{a_1 a_2 + b_1 b_2 + c_1 c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}}$

46. $\cos \theta = \frac{a_1 a_2 + b_1 b_2 + c_1 c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}}$

47. Take $\bar{a} \cdot \bar{b} = 0$

48. $W = \bar{F} \cdot \bar{AB}$

49. Work done = 5 $\Rightarrow \bar{F} \cdot \bar{S} = 5 \Rightarrow a = 3$



LEVEL - II (C.W)

SCALAR (OR) DOT PRODUCT OF VECTORS GEOMETRICAL INTERPRETATION ORTHOGONAL PROJECTION

1. If $\bar{a} = 2\bar{i} + \bar{j} + 2\bar{k}$, $\bar{b} = 5\bar{i} - 3\bar{j} + \bar{k}$, then the length of the component vector of $\bar{b}$ perpendicular to $\bar{a}$ is

1) $\sqrt{13}$ 2) $\sqrt{26}$ 3) $\sqrt{18}$ 4) $\sqrt{20}$

2. A Parallelogram is constructed with $\bar{a}$ and $\bar{b}$ as adjacent sides such that $|\bar{a}| = a$ and $|\bar{b}| = b$. The vector which coincides with the altitude of the parallelogram and is perpendicular to the vector $\bar{a}$ is.

1) $\bar{b} - \frac{(\bar{a} \cdot \bar{b})\bar{a}}{a^2}$ 2) $\bar{a} - \frac{(\bar{a} \cdot \bar{b})\bar{b}}{b^2}$

3) $\bar{a} - \frac{(\bar{a} \cdot \bar{b})\bar{b}}{b^2}$ 4) $\bar{b} - \frac{(\bar{a} \cdot \bar{b})\bar{a}}{a^2}$

3. If $\bar{a}, \bar{b}, \bar{c}$ are position vectors of the non-collinear points A, B, C respectively, the shortest distance of A from BC is

1) $\bar{a} \cdot (\bar{b} - \bar{c})$

2) $\bar{b} \cdot (\bar{c} - \bar{a})$

3) $|\bar{b} - \bar{a}|$ 4) $\sqrt{|\bar{b} - \bar{a}|^2 - \left(\frac{(\bar{a} - \bar{b}) \cdot (\bar{c} - \bar{b})}{|\bar{c} - \bar{b}|} \right)^2}$

PROPERTIES OF DOT PRODUCT

4. If $\bar{a}$ and $\bar{b}$ are non-collinear unit vectors and $|\bar{a} + \bar{b}| = \sqrt{3}$ then $(2\bar{a} + 5\bar{b}) \cdot (3\bar{a} - \bar{b}) =$

1) $\frac{15}{4}$ 2) $\frac{15}{2}$ 3) 15 4) 16

5. If $\bar{a}, \bar{b}$ and $\bar{c}$ are perpendicular to $\bar{b} + \bar{c}, \bar{c} + \bar{a}$ and $\bar{a} + \bar{b}$ respectively and if $|\bar{a} + \bar{b}| = 6$, $|\bar{b} + \bar{c}| = 8$ and $|\bar{c} + \bar{a}| = 10$, then $|\bar{a} + \bar{b} + \bar{c}|$ is equal to

1) $5\sqrt{2}$ 2) 50 3) $10\sqrt{2}$ 4) 10

6. If $\bar{b} = 4\bar{i} + 3\bar{j}$ and $\bar{c}$ are two vectors perpendicular to each other in the XY plane, the vector in the same plane having components 1, 2 along $\bar{b}$ and $\bar{c}$ respectively is

1) $(-2\bar{i} + 11\bar{j})/5$ 2) $(2\bar{i} + 11\bar{j})/5$
3) $(-2\bar{i} - 11\bar{j})/5$ 4) $(2\bar{i} - 11\bar{j})/5$

7. The perpendicular distance of a corner of unit cube from a diagonal not passing through it is

1) $\sqrt{2}/3$ 2) $2/3$ 3) $1/3$ 4) 1

8. If A, B are two points on the curve $y = x^2$ in the xoy plane satisfying $\overline{OA} \cdot \bar{i} = 1$ and $\overline{OB} \cdot \bar{i} = -2$ then the length of the vector $2\overline{OA} - 3\overline{OB}$ is

1) $\sqrt{14}$ 2) $2\sqrt{51}$ 3) $3\sqrt{41}$ 4) $2\sqrt{41}$

9. $\bar{a}$ & $\bar{b}$ are two non-collinear vectors then the

value of $\left\{ \frac{\bar{a}}{|\bar{a}|^2} - \frac{\bar{b}}{|\bar{b}|^2} \right\}^2$ is equal to

1) $\frac{\bar{a} - \bar{b}}{ab}$ 2) $\left(\frac{\bar{a} - \bar{b}}{ab} \right)^2$

3) $\left(\frac{\bar{a} - \bar{b}}{ab} \right)^3$ 4) $(\bar{a} - \bar{b})^2$

10. Let $\vec{u} = \vec{i} + \vec{j}$, $\vec{v} = \vec{i} - \vec{j}$ and $\vec{w} = \vec{i} + 2\vec{j} + 3\vec{k}$. If $\vec{n}$ is a unit vector such that $\vec{u} \cdot \vec{n} = 0$ and

$$\vec{v} \cdot \vec{n} = 0, \text{ then } |\vec{w} \cdot \vec{n}| = \quad (\text{AIE-2003})$$

- 1) 1 2) 2 3) 3 4) 0

11. If a, b, c are the $p^{\text{th}}, q^{\text{th}}, r^{\text{th}}$ terms of an HP and

$$\vec{u} = (q-r)\vec{i} + (r-p)\vec{j} + (p-q)\vec{k}$$

$$\vec{v} = \frac{\vec{i}}{a} + \frac{\vec{j}}{b} + \frac{\vec{k}}{c} \text{ then}$$

- 1) $\vec{u}, \vec{v}$ are parallel vectors

- 2) $\vec{u}, \vec{v}$ are orthogonal vectors

- 3) $\vec{u} \cdot \vec{v} = 1$

- 4) $\vec{u} \times \vec{v} = \vec{i} + \vec{j} + \vec{k}$

12. If a parallelogram is constructed on the vectors

$$\vec{a} = 3\vec{p} - \vec{q}, \vec{b} = \vec{p} + 3\vec{q} \text{ and } |\vec{p}| = |\vec{q}| = 2$$

and angle between $\vec{p}$ and $\vec{q}$ is $\frac{\pi}{3}$, then the

ratio of the lengths of the sides is

- 1) $\sqrt{7} : \sqrt{13}$

- 2) $\sqrt{6} : \sqrt{2}$

- 3) $\sqrt{3} : \sqrt{5}$

- 4) 1 : 2

SCALAR (DOT) PRODUCT ANGLE BETWEEN TWO VECTORS

13. If $\vec{a} = 2\vec{m} + \vec{n}, \vec{b} = \vec{m} - 2\vec{n}$, Angle between the unit vectors $\vec{m}$ and $\vec{n}$ is 60° . a, b are the sides of a parallelogram, then the lengths of the diagonals are

- 1) $\sqrt{7}, \sqrt{5}$

- 2) $\sqrt{13}, \sqrt{5}$

- 3) $\sqrt{7}, \sqrt{13}$

- 4) $\sqrt{11}, \sqrt{13}$

14. In a right angled $\triangle ABC$. the hypotenuse $AB = p$ then $AB \cdot AC + BC \cdot BA + CA \cdot CB$ is

- 1) $2p^2$ 2) $p^2/2$ 3) p^2 4) 0

15. The triangle ABC is defined by the vertices

$$A = (0, 7, 10), B = (-1, 6, 6) \text{ and } C = (-4, 9, 6).$$

Let D be the foot of the altitude from B to the side AC. Then $\vec{DB} =$

- 1) $\vec{i} + 2\vec{j} + 2\vec{k}$

- 2) $\vec{i} - 2\vec{j} - 2\vec{k}$

- 3) $\vec{i} + 2\vec{j} - 2\vec{k}$

- 4) $\vec{i} - 2\vec{j} + 2\vec{k}$

VECTOR EQUATION OF A PLANE-NORMAL FORM

16. A plane is at a distance of 8 units from the origin and is perpendicular to the vector

$2\vec{i} + \vec{j} + 2\vec{k}$ then the equation of the plane is

- 1) $\vec{r} \cdot (2\vec{i} + \vec{j} + 2\vec{k}) = 8$ 2) $\vec{r} \cdot (\vec{i} + 2\vec{j} + 2\vec{k}) = 24$

- 3) $\vec{r} \cdot (2\vec{i} + 2\vec{j} + \vec{k}) = 24$ 4) $\vec{r} \cdot (2\vec{i} + \vec{j} + 2\vec{k}) = 24$

17. The angle between the planes passing through the points A(0, 0, 0), B(1, 1, 1), C(3, 2, 1) & the planes passing through A(0, 0, 0), B(1, 1, 1), D(3, 1, 2) is

- 1) 90° 2) 45° 3) 120° 4) 30°

APPLICATION OF SCALAR PRODUCTS IN MECHANICS TO FIND THE WORK DONE

18. The point of application of the force (-2, 4, 7) is displaced from the point (3, -5, 1) to the point (5, 9, 7). But the force is suddenly halved when the point of application moves half the distance. The work done by the force is

- 1) 70 2) 70.5 3) 75 4) 75.5

19. If forces of magnitudes 6 and 7 units acting in the directions $\vec{i} - 2\vec{j} + 2\vec{k}$ and $2\vec{i} - 3\vec{j} - 6\vec{k}$ respectively act on a particle which is displaced from the point P(2, -1, -3) to Q(5, -1, 1), then the work done by the forces is

- 1) 4 units 2) -4 units 3) 7 units 4) -7 units

20. A constant force $3\vec{i} + 4\vec{j} - 5\vec{k}$ acts on a particle at $\vec{i} + 2\vec{j} + 2\vec{k}$ and moves it to a point on the z-axis which is 3 units from origin, the work done is

- 1) 16 2) -16 3) 14 4) -14

LEVEL-II (C.W)-KEY

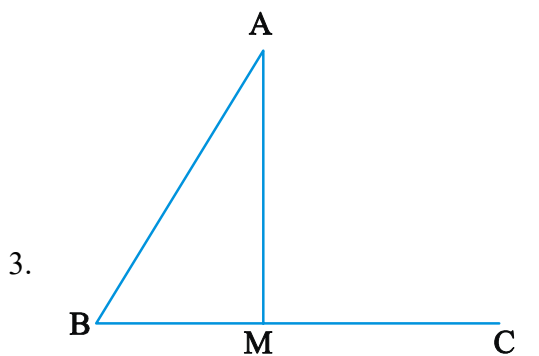
- 01) 2 02) 1 03) 4 04) 2 05) 4 06) 1
07) 1 08) 4 09) 2 10) 3 11) 2 12) 1
13) 3 14) 3 15) 2 16) 4 17) 3 18) 2
19) 1 20) 2

LEVEL-II (C.W)-HINTS

1. $\vec{b} - \frac{(\vec{a} \cdot \vec{b})\vec{a}}{|\vec{a}|^2}$

2. Required vector = Component vector of $\vec{b}$

$$\text{perpendicular to } \vec{a} = \vec{b} - \frac{(\vec{b} \cdot \vec{a})\vec{a}}{|\vec{a}|^2}$$



$$BM = \frac{\overline{BA} \cdot \overline{BC}}{|\overline{BC}|}, \quad AM = \sqrt{AB^2 - BM^2}$$

4. $|\vec{a} + \vec{b}| = \sqrt{3} \Rightarrow \vec{a} \cdot \vec{b} = \frac{1}{2}$

Find $(2\vec{a} + 5\vec{b}) \cdot (3\vec{a} - \vec{b}) = \frac{15}{2}$

5. $|\vec{a} + \vec{b}|^2 + |\vec{b} + \vec{c}|^2 + |\vec{c} + \vec{a}|^2 = 200$
 $\Rightarrow 2(|\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2) = 200 (\because \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = 0)$

$$|\vec{a} + \vec{b} + \vec{c}| = \sqrt{100} = 10$$

6. Let $\vec{d} = x\vec{i} + y\vec{j}$, and $\vec{c} = -3\vec{i} + 4\vec{j}$

$$\frac{\vec{d} \cdot \vec{b}}{|\vec{b}|} = 1 \Rightarrow 4x + 3y = 5, \quad \frac{\vec{d} \cdot \vec{c}}{|\vec{c}|} = 2$$

$$\Rightarrow -3x + 4y = 10$$

Solve these two equations.

7. The perpendicular distance of a corner of unit cube

from a diagonal not passing through it is $\sqrt{\frac{2}{3}}$.

8. Let $A(x_1, y_1)$ & $B(x_2, y_2)$ lie on the parabola so that $\vec{OA} = \vec{i} + \vec{j}$, $\vec{OB} = -2\vec{i} + 4\vec{j}$

9. Conceptual

10. Let $\vec{n} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$

$$\therefore \vec{u} \cdot \vec{n} = 0 \Rightarrow a_1 + a_2 = 0$$

$$\vec{v} \cdot \vec{n} = 0 \Rightarrow a_1 - a_2 = 0$$

$$\Rightarrow a_1 = 0, a_2 = 0$$

Also, $a_3 = 1$ ($\because \vec{n}$ is a unit vector)

$$\therefore |\vec{w} \cdot \vec{n}| = |\vec{w}| |\vec{n}| = 3.$$

11. Use $\frac{1}{a} = A + (P-1)D$ & $q-r = \frac{1}{b} - \frac{1}{c} = \frac{c-b}{bc}$ etc

12. Find $|\vec{a}| = \sqrt{9p^2 + q^2 - 6pq \cos 60^\circ} = 2\sqrt{7}$;

$$|\vec{b}| = \sqrt{p^2 + 9q^2 + 6pq \cos 60^\circ} = 2\sqrt{13}$$

13. Length of the diagonals are $|\vec{a} + \vec{b}|, |\vec{a} - \vec{b}|$

14. $\overline{AB} \cdot \overline{AC} + \overline{BC} \cdot \overline{BA} + \overline{CA} \cdot \overline{CB}$
expand and simplify.

15. Check from options that $\overline{AC} \cdot \overline{BD} = 0$

16. $\vec{r} \cdot \hat{n} = p \Rightarrow \vec{r} \cdot \left(\frac{2\vec{i} + \vec{j} + 2\vec{k}}{3} \right) = 8$
 $\Rightarrow \vec{r} \cdot (2\vec{i} + \vec{j} + 2\vec{k}) = 24$

17. Plane through ABC is $-x + 2y - z = 0$

plane through ABD is $x + y + 2z = 0$

$$\vec{n}_1 = -\vec{i} + 2\vec{j} - \vec{k}, \quad \vec{n}_2 = \vec{i} + \vec{j} + 2\vec{k}$$

$$\cos \theta = \frac{\vec{n}_1 \cdot \vec{n}_2}{|\vec{n}_1| |\vec{n}_2|} = \frac{1}{2} \Rightarrow \theta = 60^\circ \text{ (or) } 120^\circ$$

18. $F \cdot \frac{\vec{d}}{2} + \frac{\vec{F}}{2} \cdot \frac{\vec{d}}{2}$

19. $\vec{F} = \vec{F}_1 + \vec{F}_2 = (4, -7, -2), \vec{d} = \overline{AB} = (3, 0, 4)$

$$W.D. = \vec{F} \cdot \vec{d} = 4$$

20. $W = \vec{F} \cdot \overline{AB}$

$$= (3\vec{i} + 4\vec{j} - 5\vec{k}) \cdot (3\vec{k} - (\vec{i} + 2\vec{j} + 2\vec{k}))$$

$$= (3\vec{i} + 4\vec{j} - 5\vec{k}) \cdot (-\vec{i} - 2\vec{j} + \vec{k}) = -16.$$



LEVEL - II (H.W)

SCALAR (OR) DOT PRODUCT OF VECTORS GEOMETRICAL INTERPRETATION: ORTHOGONAL PROJECTION:

1. The vector component of $\vec{i} + \vec{j} + \vec{k}$

perpendicular to the vector $2\vec{i} - \vec{j} + 2\vec{k}$ is

1) $\frac{\vec{i} + 4\vec{j} + \vec{k}}{3}$

2) $\frac{\vec{i} - 4\vec{j} - 3\vec{k}}{4}$

3) $\vec{i} + \vec{j} - \frac{1}{2}\vec{k}$

4) $\vec{0}$

2. If $\vec{a} = 3\vec{i} + \vec{j} + 2\vec{k}$ and $\vec{b} = \vec{i} + 2\vec{j} + 3\vec{k}$ then a unit vector in the direction of the resultant of orthogonal projection of $\vec{b}$ on $\vec{a}$ and the projection of $\vec{b}$ on a line perpendicular to $\vec{a}$ is

$$\begin{array}{ll} 1) \frac{\vec{i} + 2\vec{j} + 3\vec{k}}{\sqrt{14}} & 2) \frac{2\vec{i} + \vec{j} + 3\vec{k}}{\sqrt{14}} \\ 3) \frac{3\vec{i} + \vec{j} + 2\vec{k}}{\sqrt{14}} & 4) \frac{\vec{i} + 3\vec{j} + 2\vec{k}}{\sqrt{14}} \end{array}$$

3. The projection of the vector $\vec{i} + \vec{j} + \vec{k}$ on the line whose vector equation is $\vec{r} = (3+t)\vec{i} + (2t-1)\vec{j} + 3t\vec{k}$, t being the scalar parameter, is

$$(1) \frac{1}{\sqrt{14}} \quad (2) 6 \quad (3) \frac{6}{\sqrt{14}} \quad (4) 10$$

PROPERTIES OF DOT PRODUCT

4. Let $\vec{u}, \vec{v}, \vec{w}$ be such that $|\vec{u}|=1, |\vec{v}|=2, |\vec{w}|=3$. If the projection of $\vec{v}$ along $\vec{u}$ is equal to that of $\vec{w}$ along $\vec{u}$ and $\vec{v}, \vec{w}$ are perpendicular to each other, then $|\vec{u} - \vec{v} + \vec{w}|$ equals

$$1) 2 \quad 2) \sqrt{7} \quad 3) \sqrt{14} \quad 4) 14$$

5. If $\vec{a} + \vec{b} + \vec{c} = 0$, then $\vec{a} \cdot \vec{a} + \vec{b} \cdot \vec{b} + 2\vec{a} \cdot \vec{b} - \vec{c} \cdot \vec{c} =$

$$\begin{array}{ll} 1) 2(\vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) & 2) 0 \\ 3) \text{Null vector} & 4) -2(\vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) \end{array}$$

6. The length of longer diagonal of the parallelogram constructed on $5\vec{a} + 2\vec{b}$ and

$$\vec{a} - 3\vec{b} \text{ if it is given } |\vec{a}| = 2\sqrt{2}; |\vec{b}| = 3, \text{ and}$$

$$(\vec{a}, \vec{b}) = \pi/4 \text{ is}$$

$$1) 15 \quad 2) \sqrt{113} \quad 3) \sqrt{593} \quad 4) \sqrt{395}$$

7. If $\vec{a}, \vec{b}, \vec{c}$ are three vectors such that each is inclined at an angle $\frac{\pi}{3}$ with the other two and $|\vec{a}|=1, |\vec{b}|=2, |\vec{c}|=3$ then the scalar product of the vectors $2\vec{a} + 3\vec{b} - 5\vec{c}$ and $4\vec{a} - 6\vec{b} + 10\vec{c}$ is

$$1) 188 \quad 2) -334 \quad 3) -522 \quad 4) -514$$

SCALAR (DOT) PRODUCT ANGLE BETWEEN TWO VECTORS:

8. If $\vec{a}, \vec{b}, \vec{c}$ are three mutually perpendicular vectors of equal magnitudes then the vector equally inclined to $\vec{a}, \vec{b}, \vec{c}$

$$\begin{array}{ll} 1) \frac{\vec{i} + \vec{j} + \vec{k}}{\sqrt{3}} & 2) \vec{a} - \vec{b} + \vec{c} \\ 3) \frac{\vec{a} - \vec{b} - \vec{c}}{\sqrt{3}} & 4) \frac{\vec{a} + \vec{b} + \vec{c}}{\sqrt{3}} \end{array}$$

9. If the position vectors of A, B, C, D are respectively

$$(-6, 1, 6), (6, -2, 3), (-2, -3, -1) \text{ and}$$

$$(-5, -9, -7) \text{ then}$$

$$\begin{array}{ll} 1) \angle BCA \text{ is a right angle} & 2) \angle CDA \text{ is a right angle} \\ 3) \angle ABD \text{ is a right angle} & 4) \angle ACD \text{ is a right angle} \end{array}$$

10. Let $A(2, -3, 0), B(2, -1, 1), C(0, 1, 4)$ be the vertices of $\triangle ABC$. Then angle between the median $\overline{BD}$ and the base $\overline{AC}$ is

$$1) 30^\circ \quad 2) 45^\circ \quad 3) 60^\circ \quad 4) 90^\circ$$

11. Consider points A, B, C, D with position vectors $7\vec{i} - 4\vec{j} + 7\vec{k}, \vec{i} - 6\vec{j} + 10\vec{k}, -\vec{i} - 3\vec{j} + 4\vec{k}$ and

$$5\vec{i} - \vec{j} + 5\vec{k} \text{ respectively. Then, ABCD is a}$$

(AIE-2003)

$$\begin{array}{ll} 1) \text{rhombus} & 2) \text{rectangle} \\ 3) \text{parallelogram but not a rhombus} & 4) \text{None of these} \end{array}$$

12. If ABCDEF is a regular hexagon, then $\overline{AB} \cdot \overline{AF} = \dots$

$$1) \frac{1}{2} BC^2 \quad 2) -\frac{1}{2} BC^2 \quad 3) \frac{1}{2} AC^2 \quad 4) -\frac{1}{2} AC^2$$

13. $|\vec{b}| = 6$, then $\vec{b} - 3\vec{c} = \lambda\vec{a}$ if $\lambda =$

$$1) -9, 3 \quad 2) -3, 6 \quad 3) 6, 3 \quad 4) -3, 4$$

14. The vectors $\vec{a}, \vec{b}, \vec{c}$ are of same length and taken pairwise they form equal angles, If

$$\vec{a} = \vec{i} + \vec{j}, \vec{b} = \vec{j} + \vec{k} \text{ then } \vec{c} =$$

$$1) \vec{i} + \vec{k} \quad 2) \vec{i} + \vec{j} + \vec{k} \quad 3) \vec{i} - \vec{j} \quad 4) -\vec{i} + \vec{j}$$

15. If $\vec{a} = -\vec{i} + \vec{j} + \vec{k}$ and $\vec{b} = 2\vec{i} + \vec{k}$, then the vector $\vec{c}$ satisfying the conditions that (i) it is coplanar with $\vec{a}$ and $\vec{b}$ (ii) it is perpendicular to $\vec{b}$ (iii) $\vec{a} \cdot \vec{c} = 7$ is

- 1) $-3\vec{i} + 5\vec{j} + 3\vec{k}$ 2) $-3\vec{i} + 5\vec{j} + 6\vec{k}$
3) $-6\vec{i} + \vec{k}$ 4) $-\vec{i} + 2\vec{j} + 2\vec{k}$

16. If $a + 2b + 3c = 4$ then the least value of $a^2 + b^2 + c^2$

- 1) $\frac{2}{7}$ 2) $\frac{3}{7}$ 3) $\frac{5}{7}$ 4) $\frac{8}{7}$

17. Angle between $\vec{a}$ & $\vec{b}$ is 120° . If $|\vec{b}| = 2|\vec{a}|$ and the vectors $\vec{a} + x\vec{b}$, $\vec{a} - \vec{b}$ are right angles. Then x is equal to

- 1) $\frac{1}{3}$ 2) $\frac{1}{5}$ 3) $\frac{2}{5}$ 4) $\frac{2}{3}$

18. If $\vec{a} = \vec{i} + \vec{j} + \vec{k}$ and $\vec{b} = \vec{i} - \vec{j}$ then the vectors $(\vec{a} \cdot \vec{i})\vec{i} + (\vec{a} \cdot \vec{j})\vec{j} + (\vec{a} \cdot \vec{k})\vec{k}$, $(\vec{b} \cdot \vec{i})\vec{i} + (\vec{b} \cdot \vec{j})\vec{j} + (\vec{b} \cdot \vec{k})\vec{k}$ and $\vec{i} + \vec{j} - 2\vec{k}$

- 1) Are mutually Perpendicular
2) Coplanar
3) Forms a Parallelopiped of volume 2 units
4) Forms a Parallelopiped of volume 3 units

19. $\vec{a}, \vec{b}, \vec{c}$ are the position vectors of the vertices of a triangle ABC, If $(\vec{a} - \vec{b}) \cdot (\vec{c} - \vec{b}) = 0$, then the position vector of circumcentre is

- 1) $\frac{(\vec{b} + \vec{c})}{2}$ 2) $\frac{(\vec{a} + \vec{b})}{2}$
3) $\frac{(\vec{a} + \vec{c})}{2}$ 4) $\frac{\vec{a} + \vec{b} + \vec{c}}{2}$

APPLICATIONS OF SCALAR PRODUCT IN MECHANICS TO FIND THE WORK DONE:

20. Constant forces $\vec{P} = 2\vec{i} - 5\vec{j} + 6\vec{k}$ and $\vec{Q} = -\vec{i} + 2\vec{j} - \vec{k}$ act on a particle, determine the work done when the particle is displaced from the point A with position vector $4\vec{i} - 3\vec{j} - 2\vec{k}$ to a point B with position vector

- $6\vec{i} + \vec{j} - 3\vec{k}$
1) 10 2) -15 3) -10 4) 15

LEVEL - II (H.W)-KEY

- 01) 1 02) 1 03) 3 04) 3 05) 2 06) 3
07) 2 08) 4 09) 1 10) 2 11) 4 12) 2
13) 1 14) 1 15) 1 16) 4 17) 3 18) 1
19) 3 20) 2

LEVEL - II (H.W)-HINTS

- Use $\vec{b} - \frac{(\vec{a} \cdot \vec{b})\vec{a}}{|\vec{a}|^2}$.
- Required vector $= \frac{\vec{b}}{|\vec{b}|}$.
- Since $\vec{r} = 3\vec{i} - \vec{j} + t(\vec{i} + 2\vec{j} + 3\vec{k})$
So, a vector parallel to the lines is $\vec{b} = \vec{i} + 2\vec{j} + 3\vec{k}$
Now, unit vector along the line is $\frac{\vec{i} + 2\vec{j} + 3\vec{k}}{\sqrt{1^2 + 2^2 + 3^2}}$ i.e. $= \frac{\vec{i} + 2\vec{j} + 3\vec{k}}{\sqrt{14}}$ the projection of $(\vec{i} + \vec{j} + \vec{k})$ on lines is $(\vec{i} + \vec{j} + \vec{k}) \cdot \frac{(\vec{i} + 2\vec{j} + 3\vec{k})}{\sqrt{14}} = \frac{1 + 2 + 3}{\sqrt{14}} = \frac{6}{\sqrt{14}}$
- $|\vec{u} - \vec{v} + \vec{w}|^2 = |\vec{u}|^2 + |\vec{v}|^2 + |\vec{w}|^2 + 2|\vec{u} \cdot \vec{w} - \vec{u} \cdot \vec{v} - \vec{v} \cdot \vec{w}|$
 $= 1 + 4 + 9 + 2|\vec{u} \cdot \vec{w} - \vec{u} \cdot \vec{v}|$
 $(\because \vec{v} \cdot \vec{w} = 0) = 14 + 2(|\vec{u}||\vec{w}|\cos\alpha - |\vec{u}||\vec{v}|\cos\beta)$
But, $|\vec{v}|\cos\beta = |\vec{w}|\cos\alpha$
 $\therefore |\vec{u} - \vec{v} + \vec{w}|^2 = 14 \Rightarrow |\vec{u} - \vec{v} + \vec{w}| = \sqrt{14}$.
- $\vec{a} + \vec{b} + \vec{c} = \vec{0} \Rightarrow \vec{a} + \vec{b} = -\vec{c}$
Take dot product on both sides with $\vec{a}$ and $\vec{b}$ then add.
- Longer diagonal $-4\vec{a} - 5\vec{b}$
its magnitude $= |-4\vec{a} - 5\vec{b}| = \sqrt{593}$.
- Given $(\vec{a}, \vec{b}) = (\vec{b}, \vec{c}) = (\vec{c}, \vec{a}) = 60^\circ$
Find $(2\vec{a} + 3\vec{b} - 5\vec{c}) \cdot (4\vec{a} - 6\vec{b} + 10\vec{c}) = -334$

8. $|\vec{a}| = |\vec{b}| = |\vec{c}|$ and $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = 0$.
9. $\therefore \angle BCA = 90^\circ$.
10. $\theta = \cos^{-1} \left(\frac{|\vec{BD} \cdot \vec{AC}|}{|\vec{BD}| |\vec{AC}|} \right)$.
11. $\vec{AB}$ and $\vec{DC}$ are not parallel.
12. Drawn the figure and see the $\angle BAF = 120^\circ \therefore \vec{AB} \cdot \vec{AF} = |\vec{AB}| |\vec{AF}| \cos 120^\circ$.
Here $AB = AF = a = BC$
13. Write $\vec{b} = \lambda \vec{a} + 3\vec{c}$ squaring, we get $\lambda^2 + 6\lambda - 27 = 0$
14. $\vec{c} = \vec{i} + \vec{k}$
15. Take $C = \left(-\frac{3}{2}, \frac{5}{2}, 3 \right)$ and verify
16. Consider the vectors $\vec{p} = a\vec{i} + b\vec{j} + c\vec{k}$ and $\vec{q} = \vec{i} + 2\vec{j} + 3\vec{k}$, $\cos \theta = \frac{\vec{p} \cdot \vec{q}}{|\vec{p}| |\vec{q}|} \leq 1$
17. $(\vec{a} + x\vec{b}) \cdot (\vec{a} - \vec{b}) = 0$
 $\Rightarrow |\vec{a}|^2 - (x-1)|\vec{a}|^2 - 4x|\vec{a}|^2 = 0$
(with $|\vec{b}| = 2|\vec{a}|$ and $(\vec{a} \cdot \vec{b}) = 120^\circ$)
 $\Rightarrow (2-5x)|\vec{a}|^2 = 0 \Rightarrow x = \frac{2}{5} (|\vec{a}| \neq 0)$
18. Verification
19. $\triangle ABC$ Right angled triangle at B
20. $\vec{W} = \vec{F} \cdot \vec{s}$, $\vec{F} = \vec{P} + \vec{Q}$, $\vec{s} = \vec{AB}$.



LEVEL - III

1. If $\sum_{i=1}^n \vec{a}_i = \vec{0}$ where $|\vec{a}_i| = 1$, for all i , then the value of $\sum_{1 \leq i < j \leq n} \vec{a}_i \cdot \vec{a}_j$ is
1) n^2 2) $-n^2$ 3) n 4) $-\frac{n}{2}$
2. The value of a for which the angle between $\vec{a} = 2a^2\vec{i} + 4a\vec{j} + \vec{k}$ and $\vec{b} = 7\vec{i} - 2\vec{j} + a\vec{k}$ is obtuse and the angle between $\vec{b}$ and z-axis is acute and less than $\frac{\pi}{6}$ is
1) Does not exist 2) Lies in $\left(0, \frac{1}{2}\right)$
3) Lies in $(-1, 1)$ 4) Lies in $(0, 1)$

3. If the position vector of a point P is $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$, where $x, y, z \in N$ and $\vec{a}$ is a vector given by $\vec{a} = \vec{i} + \vec{j} + \vec{k}$, then the total number of possible positions of point P for which $\vec{r} \cdot \vec{a} = 10$
1) 36 2) 72 3) 66 4) 100
4. Let $\vec{a} = 2\vec{i} + \vec{j} + \vec{k}$, $\vec{b} = \vec{i} + 2\vec{j} - \vec{k}$ and a unit vector $\vec{c}$ be coplanar. If $\vec{c}$ is perpendicular to $\vec{a}$, then $\vec{c}$ is
1) $\frac{1}{\sqrt{2}}(-\vec{j} + \vec{k})$ 2) $\frac{1}{\sqrt{3}}(-\vec{i} - \vec{j} - \vec{k})$
3) $\frac{1}{\sqrt{5}}(\vec{i} - 2\vec{j})$ 4) $\frac{1}{\sqrt{5}}(\vec{i} - \vec{j} - \vec{k})$
5. The position vector of the foot of the perpendicular from $(1, -2, -3)$ to the line $\vec{r} = \vec{i} + \vec{j} + \lambda(2\vec{i} + \vec{j} + \vec{k})$ is
1) $-2\vec{i} - \vec{j} + \vec{k}$ 2) $-\vec{i} - \vec{k}$
3) $\frac{\vec{j} + \vec{k}}{2}$ 4) $\vec{i} + \vec{j} + \vec{k}$
6. Let $\vec{a} = 2\vec{i} - \vec{j} + \vec{k}$, $\vec{b} = \vec{i} + 2\vec{j} - \vec{k}$ and $\vec{c} = \vec{i} + \vec{j} - 2\vec{k}$ be three vectors. A vector in the plane of $\vec{b}$ and $\vec{c}$ whose projection on $\vec{a}$ is of magnitude $\sqrt{\frac{2}{3}}$ is
1) $2\vec{i} - 3\vec{j} - 3\vec{k}$ 2) $2\vec{i} + 3\vec{j} + 3\vec{k}$
3) $-2\vec{i} - \vec{j} + 5\vec{k}$ 4) $2\vec{i} + \vec{j} + 5\vec{k}$
7. The $\triangle ABC$ is defined by the vertices $A(1, -2, 2)$, $B(1, 4, 0)$ and $C(-4, 1, 1)$. Let M be the foot of the altitude drawn from the vertex B to side AC. Then $\vec{BM} =$
1) $\left(\frac{-20}{7}, \frac{-30}{7}, \frac{10}{7} \right)$ 2) $(-20, -30, 10)$
3) $(2, 3, -1)$ 4) $(1, 2, 3)$

8. If p^{th}, q^{th}, r^{th} terms of a G.P. are the positive numbers a, b, c then angle between the vectors $\log a^3 \vec{i} + \log b^3 \vec{j} + \log c^3 \vec{k}$ and $(q-r)\vec{i} + (r-p)\vec{j} + (p-q)\vec{k}$ is

1) $\frac{\pi}{6}$ 2) $\frac{\pi}{2}$ 3) $\frac{\pi}{3}$ 4) $\sin^{-1}\left(\frac{1}{\sqrt{a^2+b^2+c^2}}\right)$

9. Let $\vec{a}, \vec{b}, \vec{c}$ be vectors of equal magnitude such that the angle between $\vec{a}$ and $\vec{b}$ is α , $\vec{b}$ and $\vec{c}$ is β and $\vec{c}$ and $\vec{a}$ is γ . Then the minimum value of $\cos \alpha + \cos \beta + \cos \gamma$ is

1) $\frac{1}{2}$ 2) $-\frac{1}{2}$ 3) $\frac{3}{2}$ 4) $-\frac{3}{2}$

10. Let $\vec{p}$ and $\vec{q}$ be the position vectors of P and Q respectively with respect to 'O' and $|\vec{p}| = p, |\vec{q}| = q$. if R, S, divides PQ Internally and externally in the ratio 2:3 respectively. If OR and OS are perpendicular, then

1) $9p^2 = 4q^2$ 2) $4p^2 = 9q^2$
3) $9p = 4q$ 4) $4p = 9q$

11. The vectors $\vec{X}$ and $\vec{Y}$ satisfy the equations $2\vec{X} + \vec{Y} = \vec{p}, \vec{X} + 2\vec{Y} = \vec{q}$ where $\vec{p} = \vec{i} + \vec{j}$ and $\vec{q} = \vec{i} - \vec{j}$. If θ is the angle between $\vec{X}$ and $\vec{Y}$ then

1) $\cos \theta = \frac{4}{5}$ 2) $\sin \theta = \frac{1}{\sqrt{2}}$
3) $\cos \theta = -\frac{4}{5}$ 4) $\cos \theta = -\frac{3}{5}$

12. $A = (2, 3, 5), B = (-1, 3, 2)$ and $C = (\lambda, 5, \mu)$ are the vertices of a triangle. If the median AM is equally inclined to the coordinates axes, then

1) $\lambda = 10, \mu = 7$ 2) $\lambda = -10, \mu = 7$
3) $\lambda = 7, \mu = 10$ 4) $\lambda = -7, \mu = -10$

13. The vectors $3\vec{a} - 5\vec{b}$ and $2\vec{a} + \vec{b}$ are mutually perpendicular and the vectors $\vec{a} + 4\vec{b}$ &

$-\vec{a} + \vec{b}$ are also mutually perpendicular. Then the acute angle between $\vec{a}$ and $\vec{b}$ is

1) $\cos^{-1}\left(\frac{19}{5\sqrt{43}}\right)$ 2) $\pi - \cos^{-1}\left(\frac{19}{5\sqrt{43}}\right)$
3) $\cos^{-1}\left(\frac{9}{5\sqrt{43}}\right)$ 4) $\pi - \cos^{-1}\left(\frac{9}{5\sqrt{43}}\right)$

14. If $\vec{a}$ is the position vector of A then the position vector of the foot of the perpendicular from A to the plane $\vec{r} \cdot \vec{b} = \vec{b} \cdot \vec{c}$ is.

1) $\vec{b} + \frac{(\vec{c} - \vec{a}) \cdot \vec{b}}{|\vec{b}|^2} \vec{b}$ 2) $\vec{a} + \frac{[(\vec{c} - \vec{a}) \cdot \vec{b}] \vec{b}}{|\vec{b}|^2}$
3) $\vec{b} + \frac{[(\vec{c} - \vec{b}) \cdot \vec{a}] \vec{b}}{|\vec{a}|^2}$ 4) $\vec{a} + \frac{[(\vec{c} - \vec{b}) \cdot \vec{a}] \vec{b}}{|\vec{a}|^2}$

15. Let $a = BC, b = CA, c = AB$ be the sides of the triangle ABC. If G is the centroid of $\triangle ABC$ such that $\vec{GB}$ and $\vec{GC}$ are inclined at an obtuse angle, then

1) $5a^2 > b^2 + c^2$ 2) $5c^2 > a^2 + b^2$
3) $5b^2 > a^2 + c^2$ 4) None of these

16. The position vector of A is $p\vec{i} + \vec{j}$. If A is rotated about O through an angle $\frac{\pi}{6}$ in anti clock wise direction. It coincides with B whose position vector $\vec{i} + q\vec{j}$. The value of p, q are

1) $\sqrt{3}, 3$ 2) $\sqrt{3}, \frac{1}{3}$
3) $\sqrt{3}, \sqrt{3}$ or $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$ 4) $\frac{1}{3}, \frac{1}{3}$

17. In $\triangle ABC, |\vec{CB}| = a, |\vec{CA}| = b, |\vec{AB}| = c$. CD is median through the vertex C. Then $\vec{CA} \cdot \vec{CD}$ equals

1) $\frac{1}{4}(3a^2 + b^2 - c^2)$ 2) $\frac{1}{4}(a^2 + 3b^2 - c^2)$
3) $\frac{1}{4}(a^2 + b^2 - 3c^2)$ 4) $\frac{1}{4}(-3a^2 + b^2 + c^2)$

18. In triangle ABC if $\overrightarrow{AB} = \frac{\overrightarrow{u}}{|\overrightarrow{u}|} - \frac{\overrightarrow{v}}{|\overrightarrow{v}|}$ and

$$\overrightarrow{AC} = \frac{2\overrightarrow{u}}{|\overrightarrow{u}|} \text{ where } |\overrightarrow{u}| \neq |\overrightarrow{v}| \text{ then}$$

1) $1 + \sum \cos 2A = 0$ 2) $\sum \cos 2A = 0$

3) $2 + \sum \cos 2A = 0$

4) $1 + \cos 2A + \cos 2B - \cos 2C = 0$

19. In a parallelogram ABCD,

$$|\overrightarrow{AB}| = a, |\overrightarrow{AD}| = b,$$

and $|\overrightarrow{AC}| = c$, then $\overrightarrow{DB} \cdot \overrightarrow{AB}$ has the value

1) $\frac{3a^2 + b^2 - c^2}{2}$ 2) $\frac{a^2 + 3b^2 - c^2}{2}$

3) $\frac{a^2 - b^2 + 3c^2}{2}$ 4) $\frac{a^2 + 3b^2 + c^2}{2}$

20. If A is (x_1, y_1) where $x_1 = 1$ on the curve

$$y = x^2 + x + 10. \text{ the tangent at A cuts the x-axis}$$

at B. The value of $\overrightarrow{OA} \cdot \overrightarrow{AB}$ is

1) -148

2) -352

3) 140

4) none of these

21. Let ABCD be a parallelogram such that

$$\overrightarrow{AB} = \overrightarrow{q}, \overrightarrow{AD} = \overrightarrow{p} \text{ and } \angle BAD \text{ be an acute}$$

angle. If $\overrightarrow{r}$ is the vector that coincides with the altitude directed from the vertex B to the side AD, then $\overrightarrow{r}$ is given by

1. $\overrightarrow{r} = 3\overrightarrow{q} - \frac{3(\overrightarrow{p} \cdot \overrightarrow{q})}{(\overrightarrow{p} \cdot \overrightarrow{p})} \overrightarrow{p}$ 2. $\overrightarrow{r} = -\overrightarrow{q} + \frac{(\overrightarrow{p} \cdot \overrightarrow{q})}{(\overrightarrow{p} \cdot \overrightarrow{p})} \overrightarrow{p}$

3. $\overrightarrow{r} = \overrightarrow{q} + \frac{(\overrightarrow{p} \cdot \overrightarrow{q})}{(\overrightarrow{p} \cdot \overrightarrow{p})} \overrightarrow{p}$ 4. $\overrightarrow{r} = -3\overrightarrow{q} + \frac{(\overrightarrow{p} \cdot \overrightarrow{q})}{(\overrightarrow{p} \cdot \overrightarrow{p})} \overrightarrow{p}$

LEVEL-III-KEY

01) 4 02) 1 03) 1 04) 1 05) 2 06) 3

07) 1 08) 2 09) 4 10) 1 11) 3 12) 3

13) 1 14) 2 15) 1 16) 3 17) 2 18) 1

19) 1 20) 1 21) 2

LEVEL-III-HINTS

1. $|\overrightarrow{a}_1 + \overrightarrow{a}_2 + \dots + \overrightarrow{a}_n|^2 = |\overrightarrow{a}_1|^2 + |\overrightarrow{a}_2|^2 + \dots + |\overrightarrow{a}_n|^2 + 2$

$$\sum_{1 \leq i < j \leq n} \overrightarrow{a}_i \cdot \overrightarrow{a}_j$$

2.

$$\overrightarrow{a} \cdot \overrightarrow{b} = |\overrightarrow{a}| |\overrightarrow{b}| \cos \theta$$

$$\overrightarrow{a} \cdot \overrightarrow{b} = |\overrightarrow{a}| |\overrightarrow{b}| \cos \theta$$

$$\overrightarrow{a} \cdot \overrightarrow{b} = |\overrightarrow{a}| |\overrightarrow{b}| \cos \theta$$

$$\overrightarrow{a} \cdot \overrightarrow{b} = |\overrightarrow{a}| |\overrightarrow{b}| \cos \theta$$

$$\overrightarrow{a} \cdot \overrightarrow{b} = |\overrightarrow{a}| |\overrightarrow{b}| \cos \theta$$

$$\overrightarrow{a} \cdot \overrightarrow{b} = |\overrightarrow{a}| |\overrightarrow{b}| \cos \theta$$

There is no value satisfying

3.

$$\text{No. of solutions} =$$

4.

Let and . But

$$\overrightarrow{a} \cdot \overrightarrow{b} = |\overrightarrow{a}| |\overrightarrow{b}| \cos \theta$$

5.

$$\overrightarrow{a} \cdot \overrightarrow{b} = |\overrightarrow{a}| |\overrightarrow{b}| \cos \theta$$

$$\text{Let, ,}$$

$$\text{D.R.'S of D.R.'S of line}$$

$$\text{Pro of D.R.'S} = 0$$

6.

A vector in the plane of and is or

Its projection on

$$\text{is}$$

$$\overrightarrow{a} \cdot \overrightarrow{b} = |\overrightarrow{a}| |\overrightarrow{b}| \cos \theta$$

$$\overrightarrow{a} \cdot \overrightarrow{b} = |\overrightarrow{a}| |\overrightarrow{b}| \cos \theta$$

DOT PRODUCT OF VECTORS

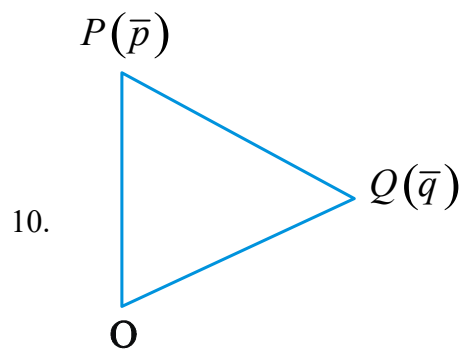
7. Here Component vector of
Perpendicular and

hence find

8. Use
etc.

9. Let
we have,

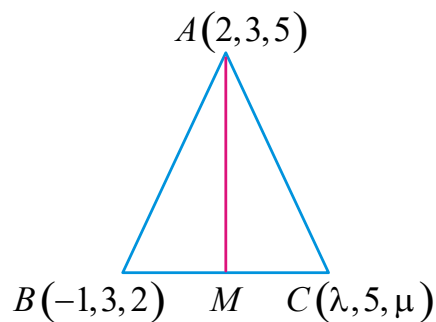
Now,



11.

JEE-MAIN-JR-MATHS VOL-I

12.


,

13.

from (1) & (2)

also

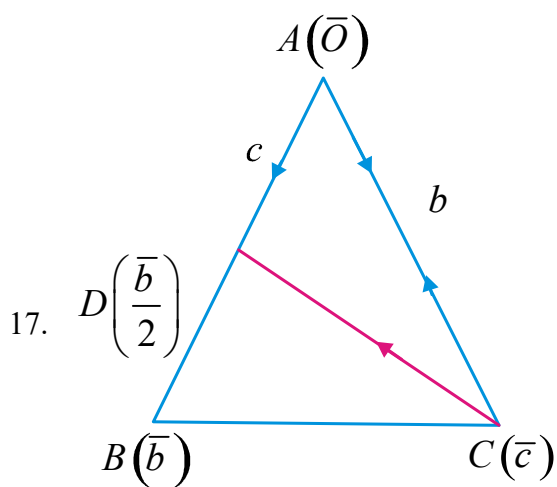
14. If then Component Vector of

along

15. Let then

 Use

16. ,



Let A be origin & then

 &

18.

 Hence

19. or

in parallelogram

20. Putting in and

 at A is 3. Tangent at A is .

it meets x-axis at B.

21. vector component of on



LEVEL - IV

1. Let $\vec{a}$ and $\vec{b}$ be mutually perpendicular unit vectors, then

I: $\vec{a} \cdot \vec{b} = 0$

II: $\vec{a}$ is equally inclined to $\vec{b}$

- 1) Both I & II are true 2) Both I & II are false
3) I is false, II is true 4) I is true, II is false

2. I: If the vectors $\vec{a} = (1, x, -2)$, $\vec{b} = (x, 3, -4)$ are mutually perpendicular, then $x = 2$

II: If $\vec{a}$ and $\vec{b}$ is perpendicular to $\vec{c}$ then $t = 5$

- 1) Only I is true 2) Only II is true
3) both I & II are true 4) neither I nor II are true

3. Arrange the value of $\vec{a} \cdot \vec{b}$ in ascending order

A: If $\vec{a}$ and $\vec{b}$ are mutually perpendicular unit vectors

B: If $\vec{a}$ and $\vec{b}$ are vectors of lengths 2, 3, 4 respectively $\vec{a}$ and $\vec{b}$ are perpendicular to $\vec{c}$ respectively

C: If $\vec{a}$ and $\vec{b}$ are vectors of lengths 4, 4, 5 respectively and $\vec{a}$ and $\vec{b}$ are perpendicular to $\vec{c}$ respectively.

- 1) A, B, C 2) C, B, A
3) B, C, A 4) B, A, C

4. A: Angle between $\vec{a}$ and $\vec{b}$ is acute angle

$\vec{a} \cdot \vec{b} > 0$

R: If $\vec{a}$ is acute then $\vec{a} \cdot \vec{b}$

- 1) Both A and R are true and R is the correct explanation of A
2) Both A and R are true and R is not correct explanation of A
3) A is true but R is false

4) A is false but R is true

5. Arrange the following angles in descending order

A: If $\vec{a}$ then $\vec{a} \cdot \vec{b}$

B: If $\vec{a}$ and $\vec{b}$ are unit vectors, then $\vec{a} \cdot \vec{b}$

C: If $\vec{a}$ and $\vec{b}$ are unit vectors then $\vec{a} \cdot \vec{b}$

- 1) A, B, C 2) C, B, A
3) B, C, A 4) B, A, C

LEVEL-IV-KEY

01) 1 02) 2 03) 1 04) 1 05) 4

LEVEL-IV-HINTS

- $\vec{a}$ and $\vec{b}$
- (I) $\vec{a}$, (II) $\vec{b}$
- (A) $\vec{a}$, (B) $\vec{b}$, (C) $\vec{c}$
- $\vec{a}$ does not lie between $\vec{b}$ and $\vec{c}$
- A) $\vec{a}$, B) $\vec{b}$, C) $\vec{c}$